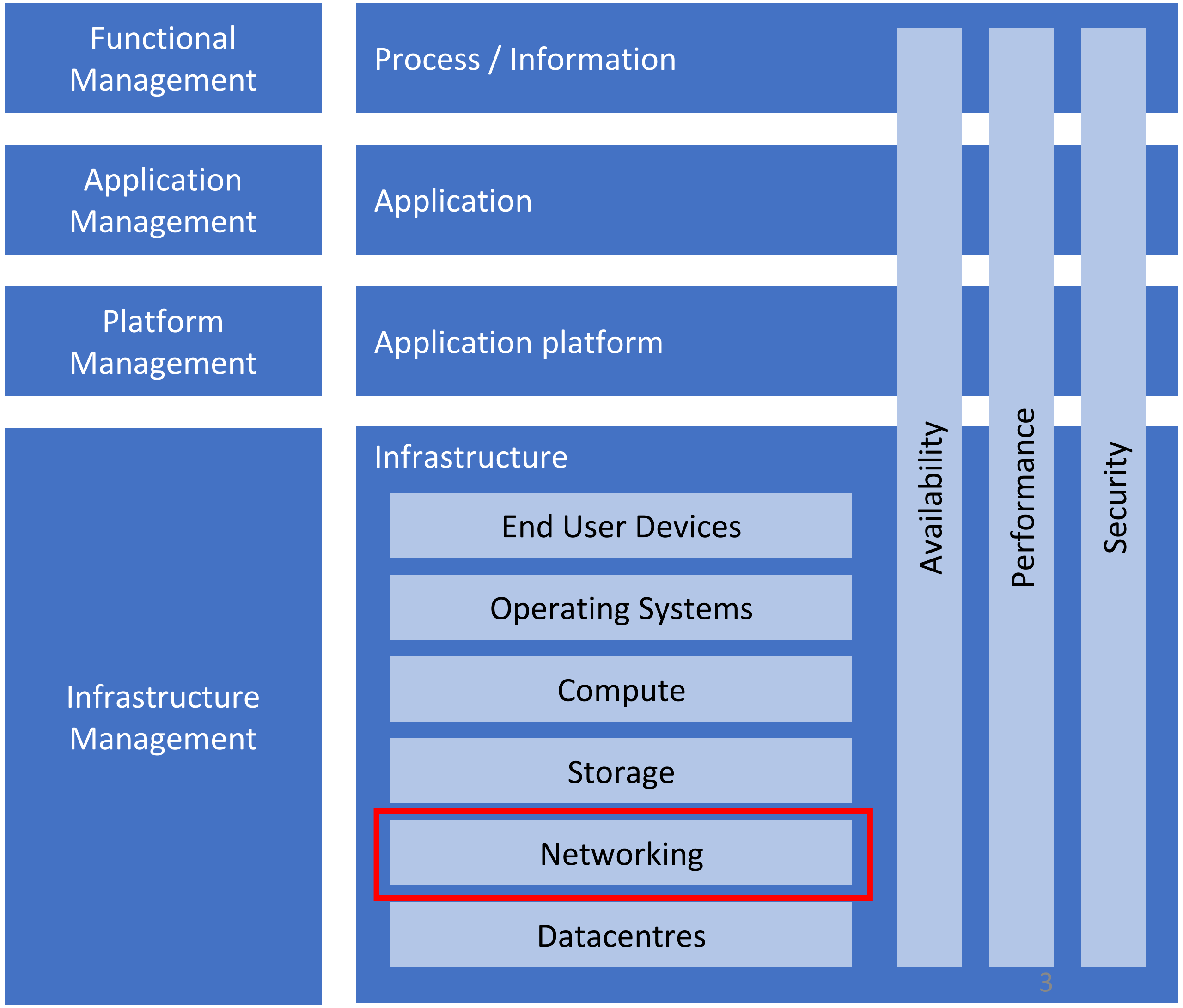
Chapter 4: Networking

Objectives

- Explain the networking building blocks
- Explain network virtualization
- Explain network availability
- Explain network performance
- Explain network security

Introduction

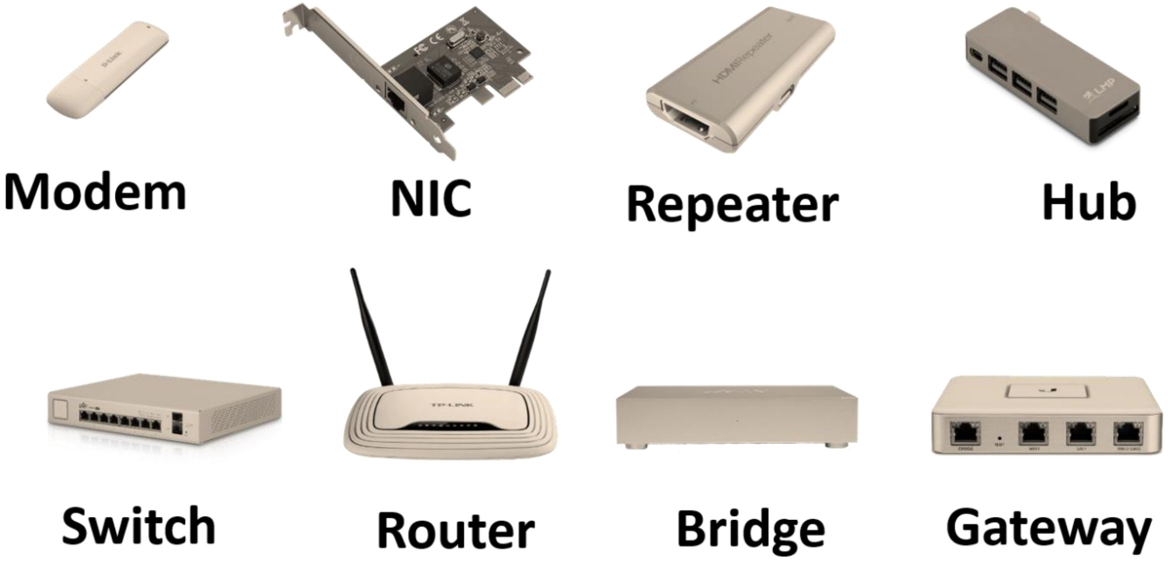
- It's the behind-the-scenes network of hardware and software that enables devices, applications, and users to connect and communicate with each other.

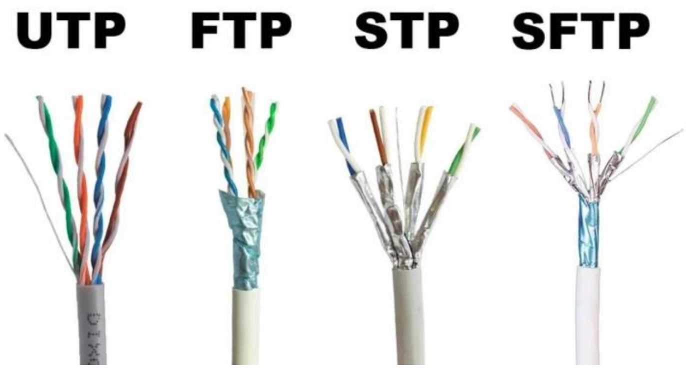


Introduction

This is a specific part of IT infrastructure that focuses on the **hardware and software** components that enable devices to **connect and communicate** with each other.

Introduction





Networking building blocks

OSI Reference Model

- The OSI Reference Model (OSI-RM) was developed in 1984 by the International Organization for Standardization (ISO)

- Seven layers define the different stages that data must go through to travel from one host to another over a network

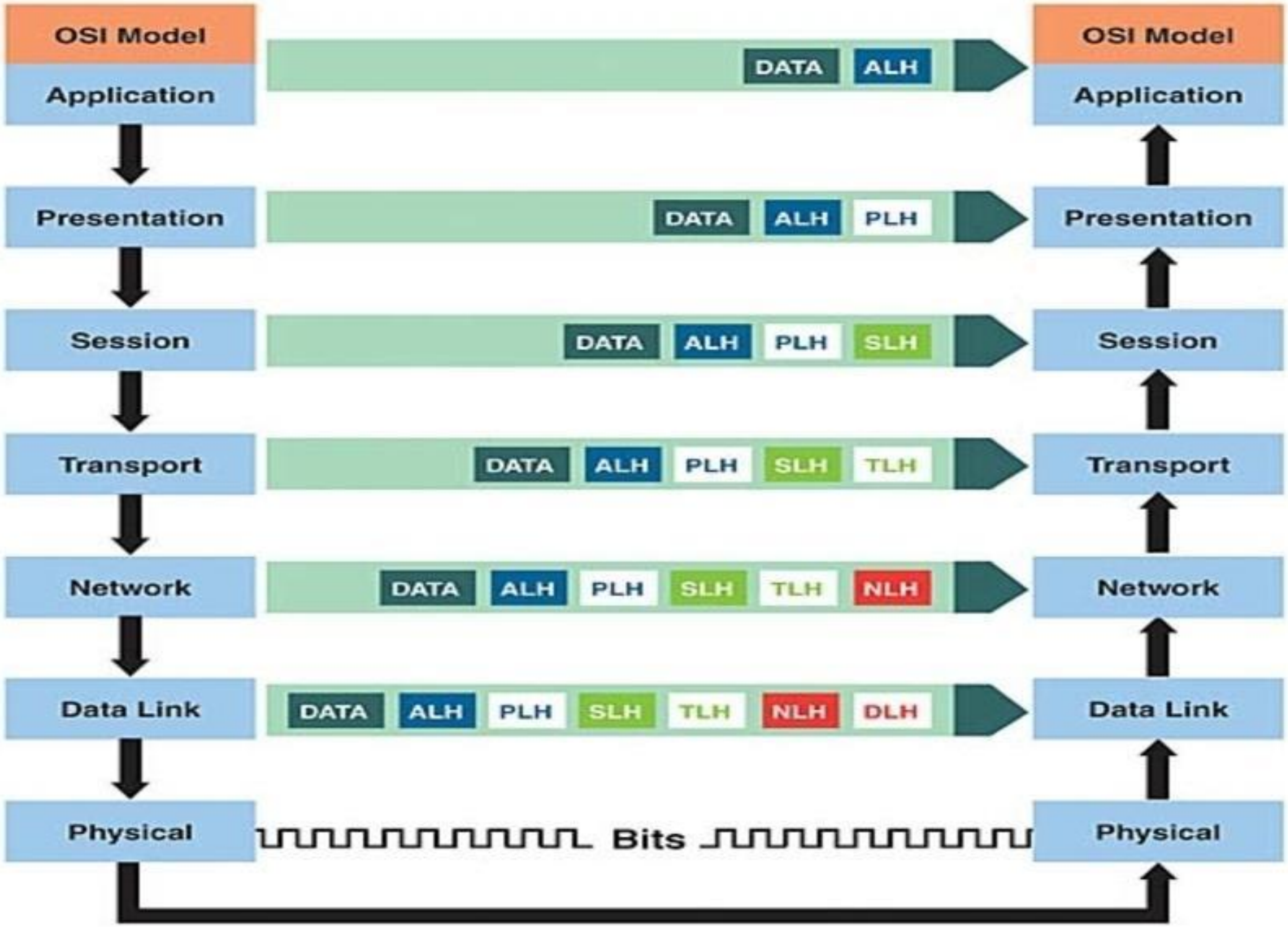
1. **P**hysical
2. **D**ata Link
3. **N**etwork
4. **T**ransport
5. **S**ession
6. **P**resentation
7. **A**pplication

OSI Reference Model

Please
Do
Not
Touch
Steve's
Pet.
Alligator

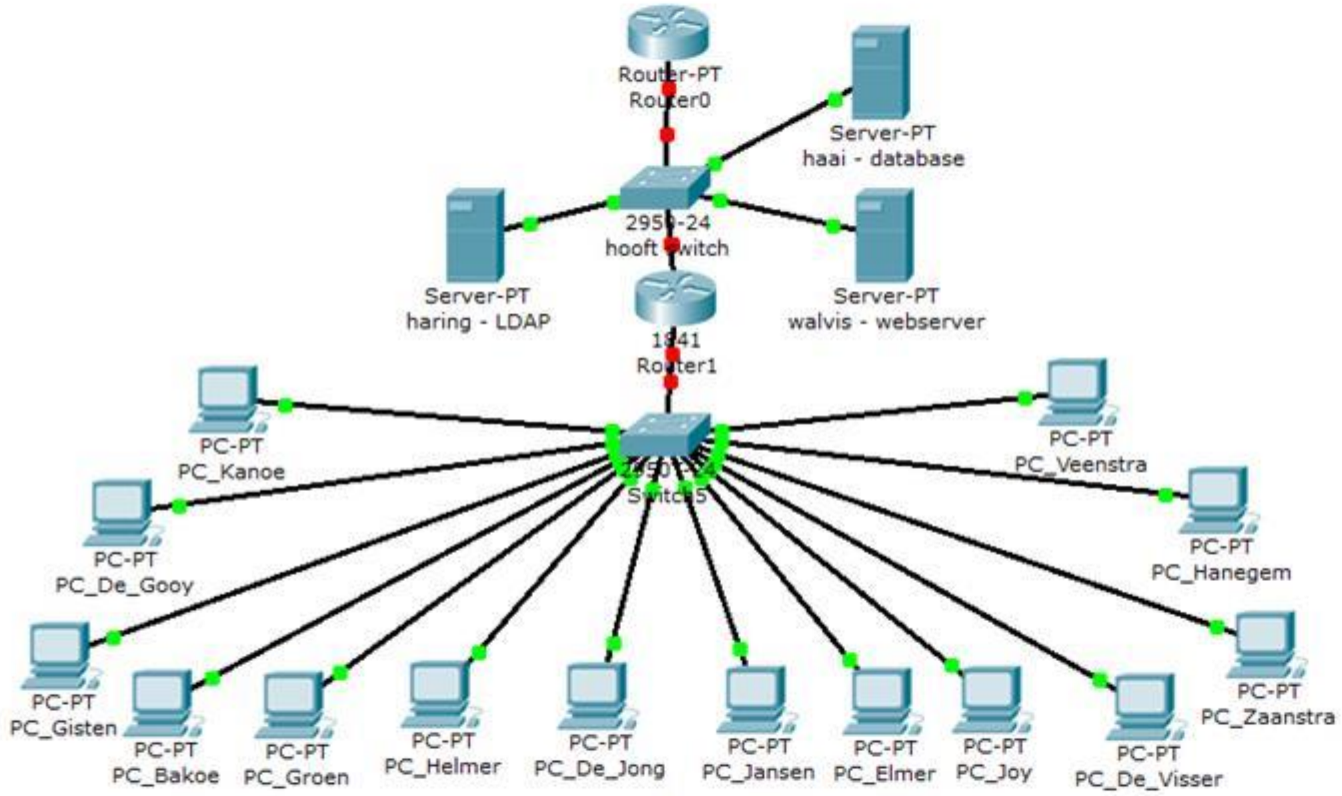
1. **P**hysical
2. **D**ata Link
3. **N**etwork
4. **T**ransport
5. **S**ession
6. **P**resentation
7. **A**pplication

OSI Reference Model



Physical layer

Physical layer

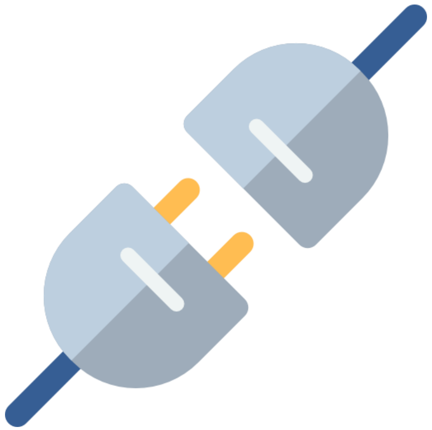


The physical layer deals with the **physical components** that **carry the data**.

Physical layer

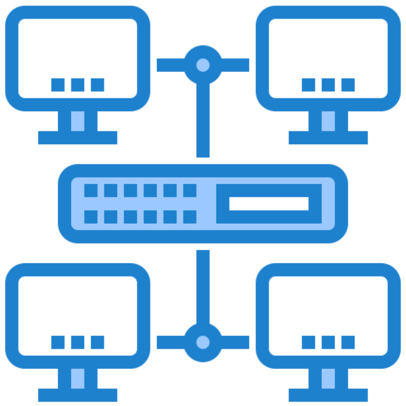
Responsibilities:

Data Transmission



Media and Connectors

Signal Characteristics



Network Topology

Physical layer

- While it might seem basic, the physical layer is crucial for everything else to function properly.

- If the cables are faulty, the connectors are dirty, or the signal strength is weak, data transmission will be disrupted impacting higher layers in the network.

Cables

The most common and suitable for server connections are:

1. Twisted Pair Cables

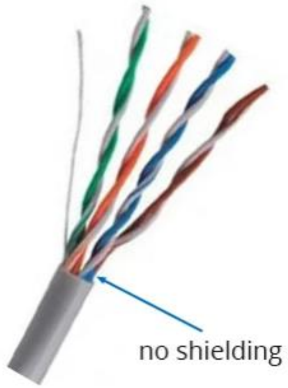
1. Fiber Optic Cables

1. Coaxial Cables

Cables - Twisted Pair Cables

Unshielded Twisted Pair (UTP)

- This is the most common type used in most home and office networks. It's relatively inexpensive and easy to install. However, it's susceptible to interference



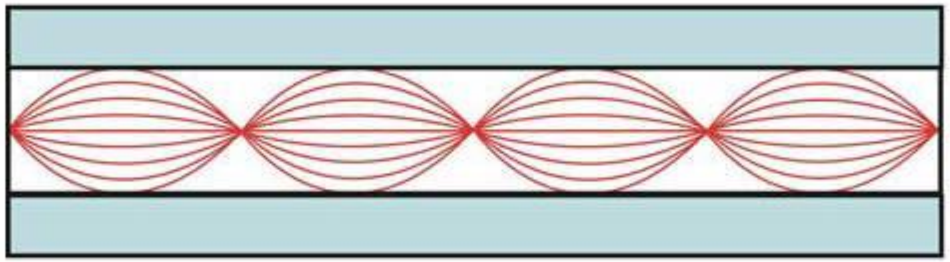
Shielded Twisted Pair (STP)

- Offers better protection against interference compared to UTP. It's typically used in environments with high electromagnetic interference

Cables - Fiber Optic Cables

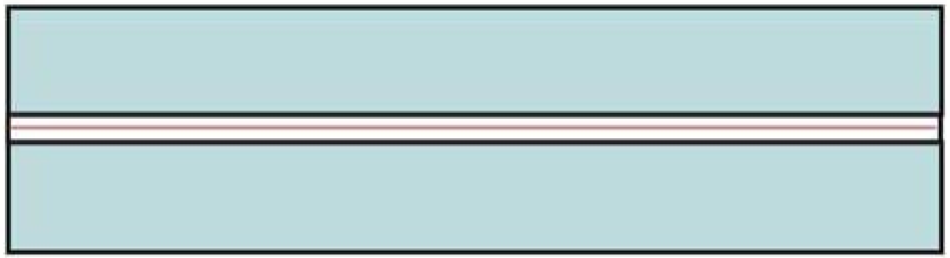
Multimode

- Uses thicker glass fibers and can transmit data over shorter distances with multiple light modes



Single-mode

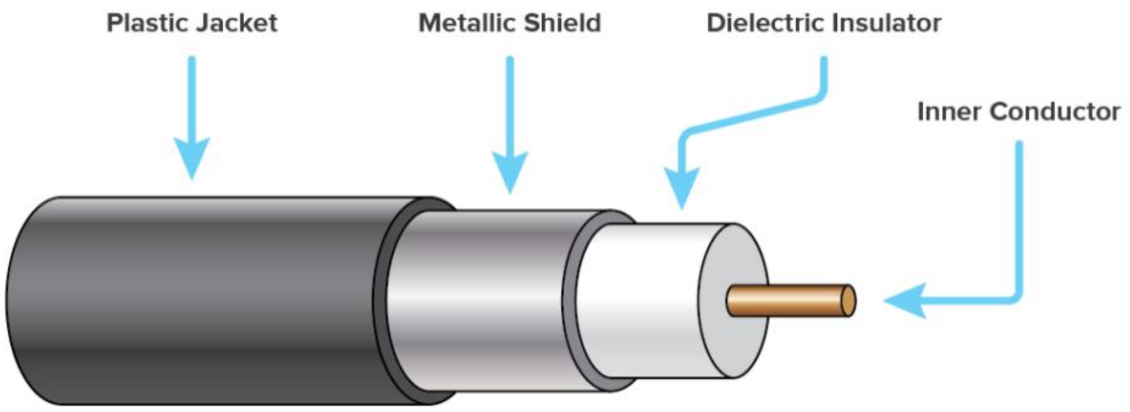
- Uses thinner glass fibers and can transmit data over longer distances with a single light mode



Cables - Coaxial Cables

- While historically used in networking, they're less common today

- They are primarily used for cable television and internet connections, not commonly used in Malaysia

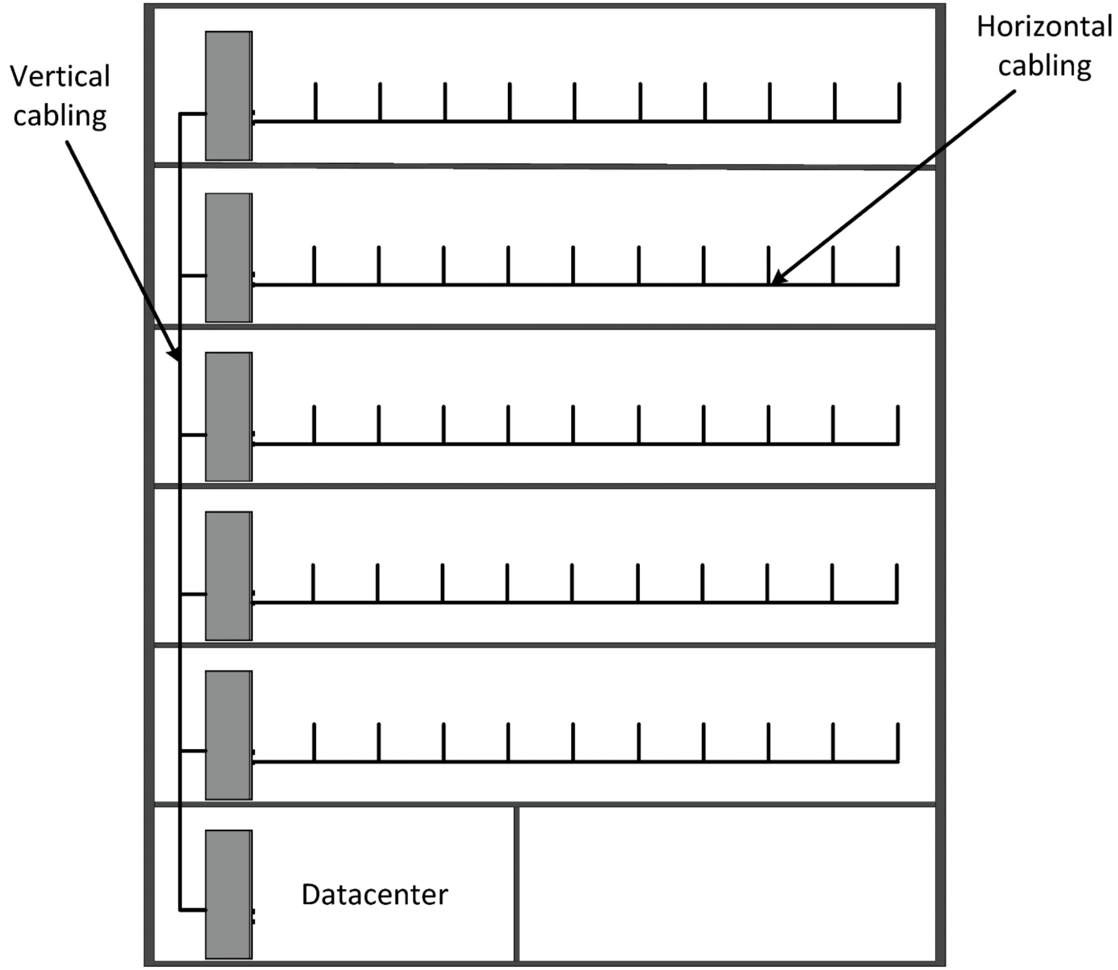


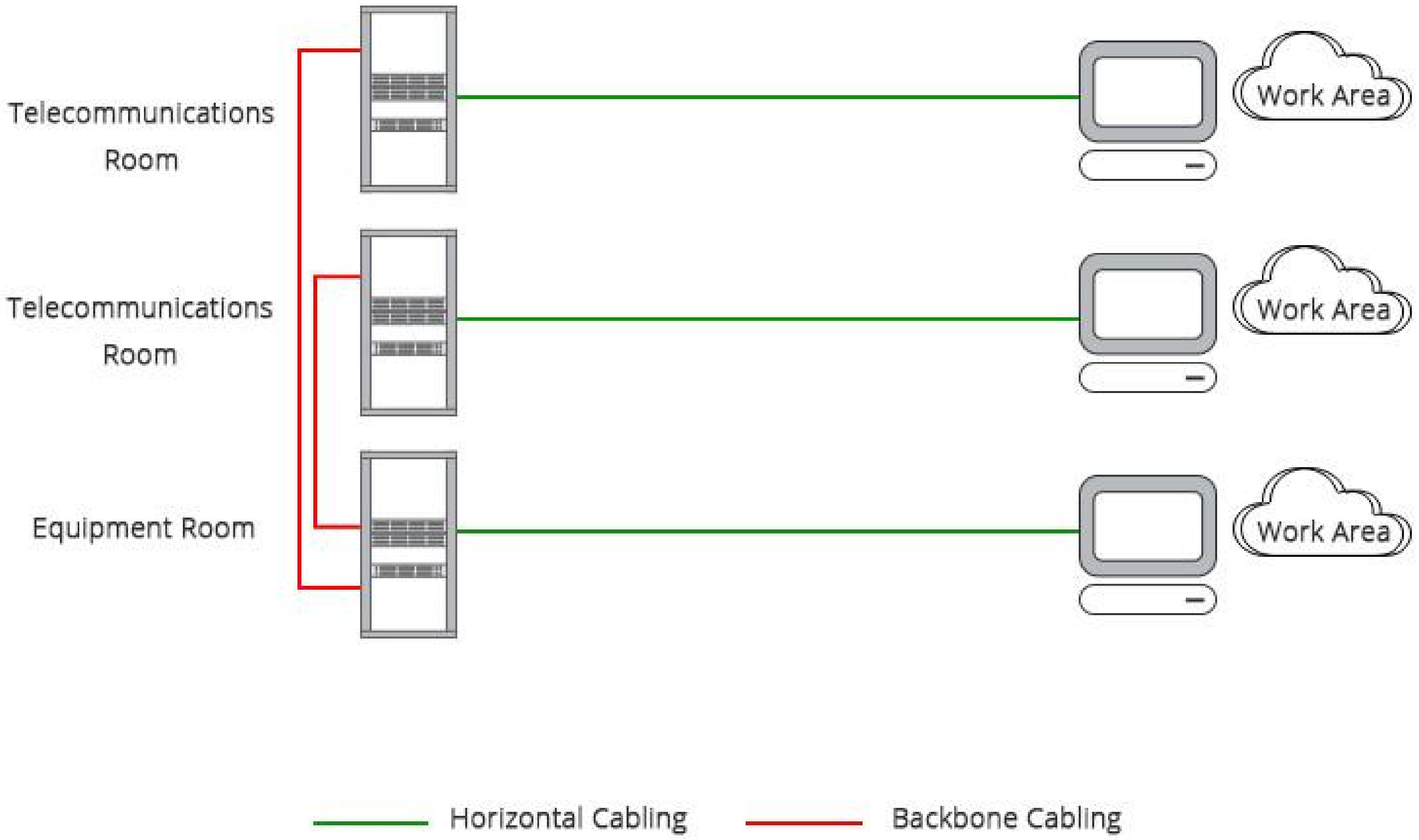
Patch panels

Vertical and horizontal cabling

- The main distribution cabling in buildings connects the patch panels on the floors to the datacentre (vertical cabling)

- Endpoints in the walls are connected to the patch panels (horizontal cabling)




Telecommunications Room
Telecommunications Room
Equipment Room
Work Area
Work Area
Work Area
Horizontal Cabling
Backbone Cabling

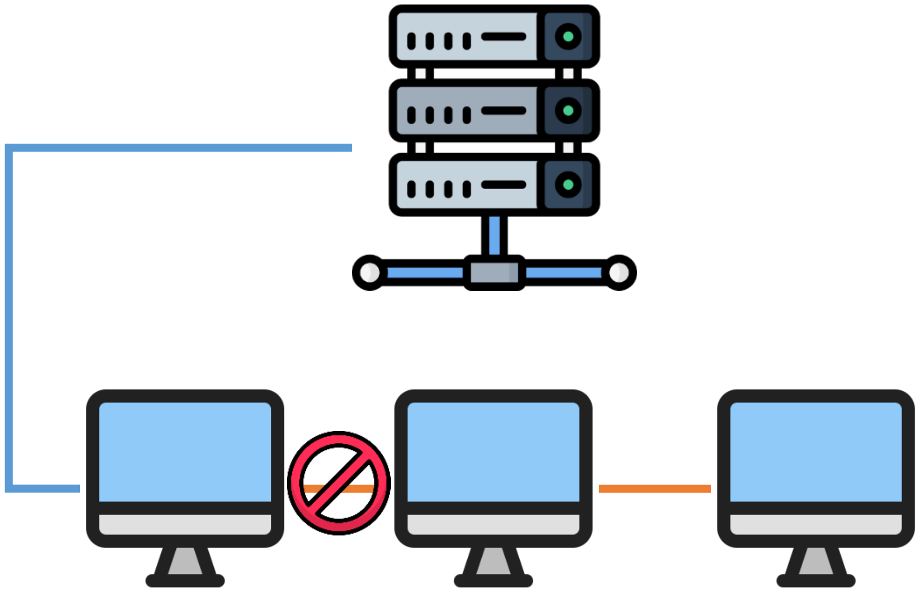
Vertical and horizontal cabling

- It is a good practice to implement redundant vertical cabling with two different physical routes, preferably using a ring layout:

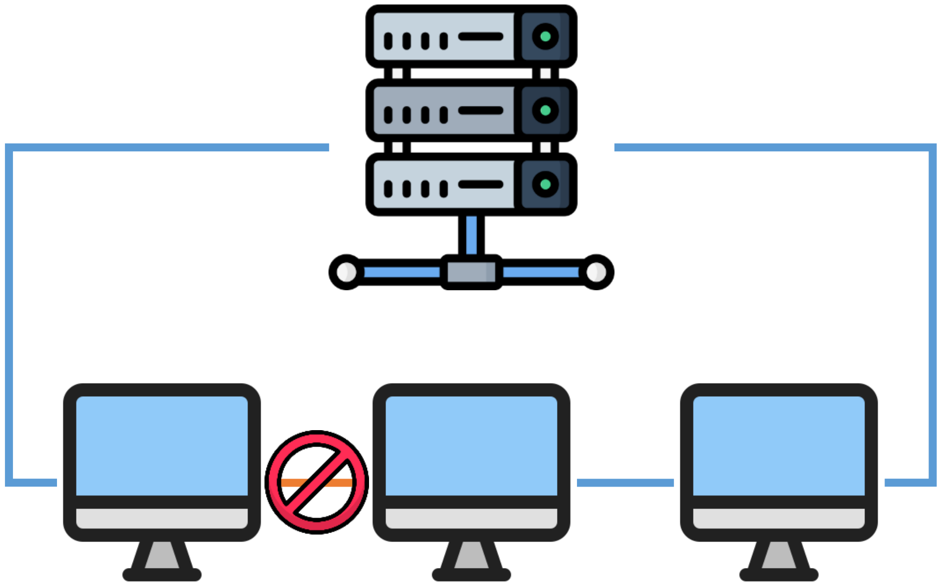
1. Enhancing Network Reliability

1. Improving Network Performance

1. Simplifying Network Management



Normal Layout



Redundant Layout

Vertical and horizontal cabling

1. Enhancing Network Reliability

- **Fault Isolation**: A failure in one cable or connection will not disrupt the entire network. This isolation prevents cascading failures and limits the impact of a single point of failure
- **Increased Uptime**: Ensures critical network services remain accessible even in the event of a cable or component failure, minimizing downtime
- **Disaster Recovery**: In case of catastrophic events like fires or floods, having multiple physical routes increases the chances of at least one path surviving

Vertical and horizontal cabling

2. Improving Network Performance

- **Load Balancing**: Distributing traffic across two physical paths can help balance network load, preventing congestion and improving overall performance

- **Reduced Latency**: Shorter cable lengths and optimized routing can lead to reduced latency and faster data transmission

Vertical and horizontal cabling

3. Simplifying Network Management

- **Centralized Management**: A ring topology can simplify network management by providing a centralized point for monitoring and control

- **Easy Troubleshooting**: Identifying and resolving network issues can be streamlined with a ring topology due to its clear structure

Leased lines

- Leased lines are dedicated data connections between two locations, provided by a telecom provider. Leased lines are based on:
  1. T (USA, Canada, and Japan) or E (most other countries) carrier lines

  1. Synchronous Optical Network (SONET) is a communication protocol designed to transmit large amounts of data over long distances using optical fiber

  1. Synchronous Digital Hierarchy (SDH) is the international standard equivalent to SONET (Synchronous Optical Network). It's a multiplexing framework used in telecommunications to transmit multiple digital data streams over a single optical fiber

  1. Dark fiber is unused optical fiber cables that have been installed but are not actively transmitting data

Leased lines

- Dark fiber:
  - Telecommunication companies often install excess fiber optic cables when building their networks to accommodate future growth
  - Businesses can lease these unused fibers from the telecommunication company
  - The lessee then installs their own equipment (transmitters, receivers, etc.) to "light up" the fiber and create their own private network

Leased lines

- Benefits of Dark Fiber
  - **High Bandwidth**: Dark fiber offers virtually unlimited bandwidth, allowing for massive data transfer speeds
  - **Low Latency**: Data travels directly between points, minimizing delays
  - **Security**: Since you control the entire network, security is significantly enhanced
  - **Scalability**: You can easily increase capacity as your needs grow
  - **Reliability**: Dark fiber is generally more reliable than shared networks due to fewer potential points of failure

Think of it as a highway: the road itself (dark fiber) can handle any speed, but the cars (equipment) determine how fast you can go.

Internet access

Types of Internet Access

1. Dedicated Internet Access (DIA): Offers high-speed, reliable connectivity with guaranteed bandwidth

1. Broadband: Provides high-speed internet access over cable, DSL, or fiber optic networks

1. Wireless Internet: Includes Wi-Fi, cellular data, and satellite internet

Datalink layer

Datalink Layer

- The data link layer, also known as layer 2 in the OSI model, acts as a bridge between the physical layer (layer 1) and the network layer (layer 3) in IT infrastructure networking.

- It's responsible for ensuring **reliable** and **error-free data transfer** between devices on the same network segment.

Datalink Layer

- The data link layer takes packets received from the network layer and breaks them down into smaller units called frames.
- These frames are manageable chunks of data suitable for transmission on the physical layer.
- Imagine the data link layer as a worker in a packaging facility, taking large boxes (packets) and dividing them into smaller packages (frames) for easier handling.

Datalink Layer

Adding Addresses

- Each frame includes the Media Access Control (MAC) addresses of the sender and receiver devices.

- These MAC addresses are unique identifiers assigned to network interfaces, similar to a home address for your device on the network.

- This addressing allows the data link layer to know where to send and receive information.

Datalink Layer

Error Detection and Correction

- Data transmission errors can occur due to electrical interference or other factors.

- The data link layer employs various techniques like checksums to detect errors in the received frames.

- In some cases, it can even request retransmission of corrupted frames to ensure data integrity.

Datalink Layer

Media Access Control (MAC)

- With multiple devices sharing a network segment, the data link layer manages how devices access the physical medium (cable or wireless) to transmit data.

- This prevents collisions and ensures orderly data flow. Imagine a group discussion where the data link layer acts as a moderator, giving each device a turn to speak (transmit) and avoiding everyone talking over each other.

PAN, LAN, MAN, WAN

- Personal Area Network
- Local Area Network
- Metropolitan Area Network
- Wide Area Network

Common Protocols

Several protocols operate at the data link layer, including:

- Ethernet: The most widely used protocol for wired LANs.

- Wi-Fi: Enables wireless communication between devices.

- Point-to-Point Protocol (PPP): Commonly used for dial-up and internet connections.

Ethernet

- Developed at Xerox PARC between 1973 and 1975
- Originally employed a shared medium topology, based on coax cable
- Later Ethernet used twisted pair cabling with hubs and switches
  - Decreased the vulnerability of the network caused by broken cables or bad connectors
- An Ethernet packet contains:
  - Source and destination MAC addresses
  - Data that needs to be transported (payload)
  - Cyclic redundancy check

Preamble	Frame delimiter	Destination MAC address	Source MAC address	Length	Payload	CRC checksum

Switching

- Switches split a single network segment into multiple segments, each segment has one device
- Switches learn which MAC address is connected to which port
- Data sent to a certain MAC address will only be forwarded to the switch port that has that MAC address connected
- On a switched network, many simultaneous data transfers can take place, in full-duplex

Public wireless networks

- Public wireless (mobile) networks are getting more popular every day
- Public wireless networks are much less reliable than private wireless networks and have lower bandwidth
- Technologies:
    - 1G and 2G: GSM, CDMA, GPRS and EDGE
    - 3G: UMTS and HSDPA
    - 4G & 5G: LTE

Network layer

The IP protocol

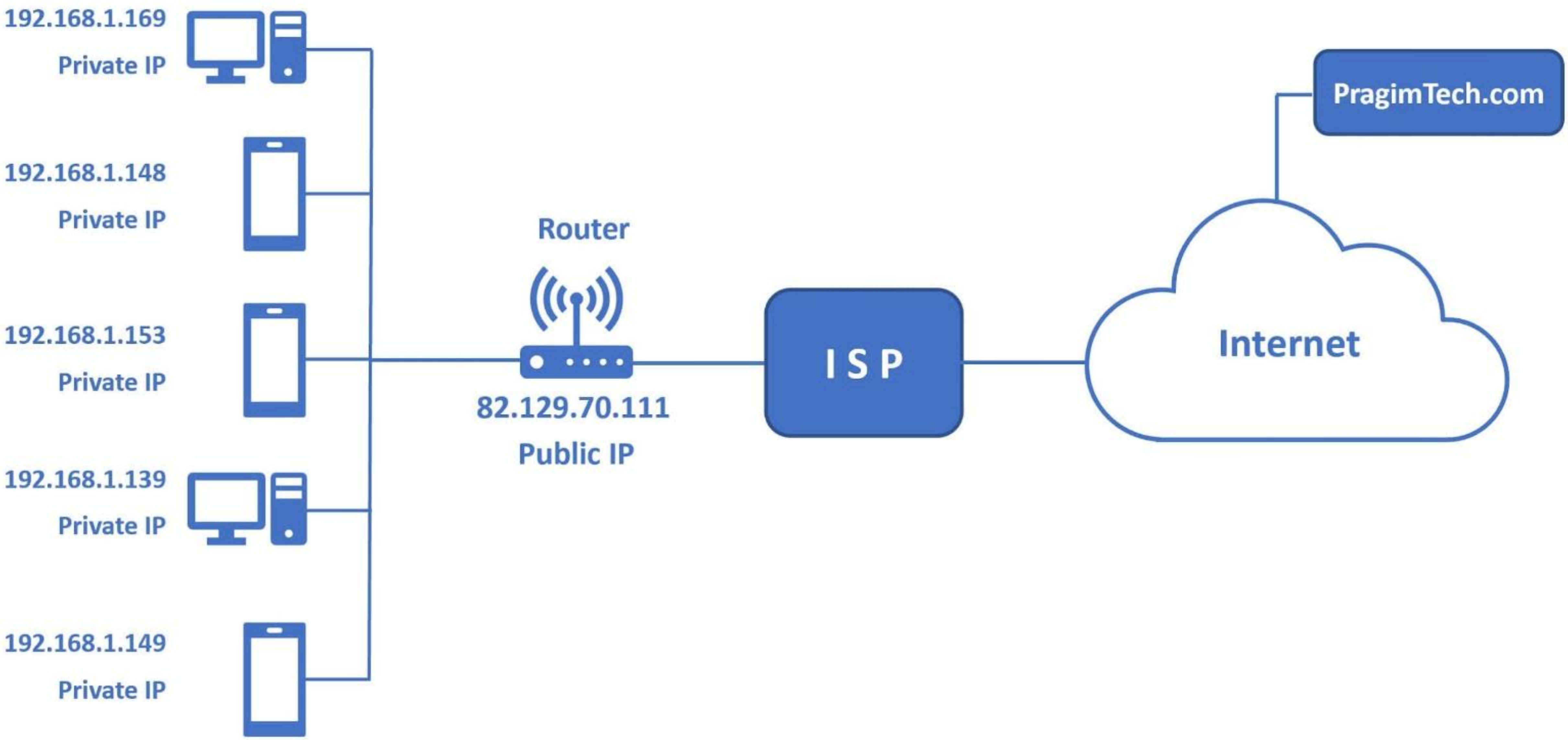
- An IP (Internet Protocol) address is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

- A server is no different, and it requires an IP address to communicate with other devices on the network.

The IP protocol

- Types of IP Addresses for Servers
  1. **Public IP Address**
     - This is a globally unique address that is visible to the entire internet.
     - It's essential for servers that need to be accessible from anywhere on the internet, such as web servers, email servers, or public-facing applications

  1. **Private IP Address**
     - These are used within a private network, such as a local area network (LAN) or a data center.
     - They are not routable on the internet and are primarily used for internal communication between devices on the network.


192.168.1.169
Private IP
192.168.1.148
Private IP
192.168.1.153
Private IP
192.168.1.139
Private IP
192.168.1.149
Private IP
Router
82.129.70.111
Public IP
I S P
Internet
PragimTech.com

The IP protocol

- **Private IP addresses** are typically assigned to local servers within a private network.

- These addresses are not routable on the internet and are used for internal communication.

- Common IP address ranges used for local servers:

  10.0.0.0 to 10.255.255.255
  172.16.0.0 to 172.31.255.255
  192.168.0.0 to 192.168.255.255

The IP protocol

- A server can have multiple IP addresses, both public and private.

- Reasons:
  - load balancing
  - hosting multiple websites
  - providing different services on the same server

The IP protocol

Static IP Address

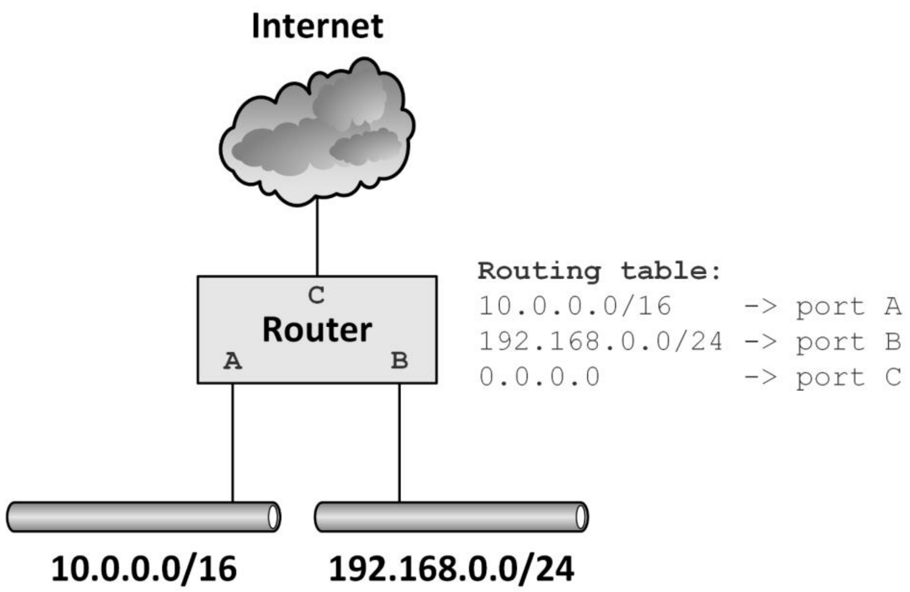
- Remains constant and doesn't change.
- It's ideal for servers that require consistent accessibility, such as web servers or email servers.

Dynamic IP Address

- Changes periodically.
- It's typically used for internal servers or devices that don't require consistent external access.

Routing

- A router copies IP packages between (sub)networks
- Routers compile routing tables to make IP packet forwarding decisions
- Routing and switching functionality may be combined in one device. A switch capable of handling routing protocols is also known as a layer 3 switch



Routing

- Routing in the context of a server typically refers to two main aspects:

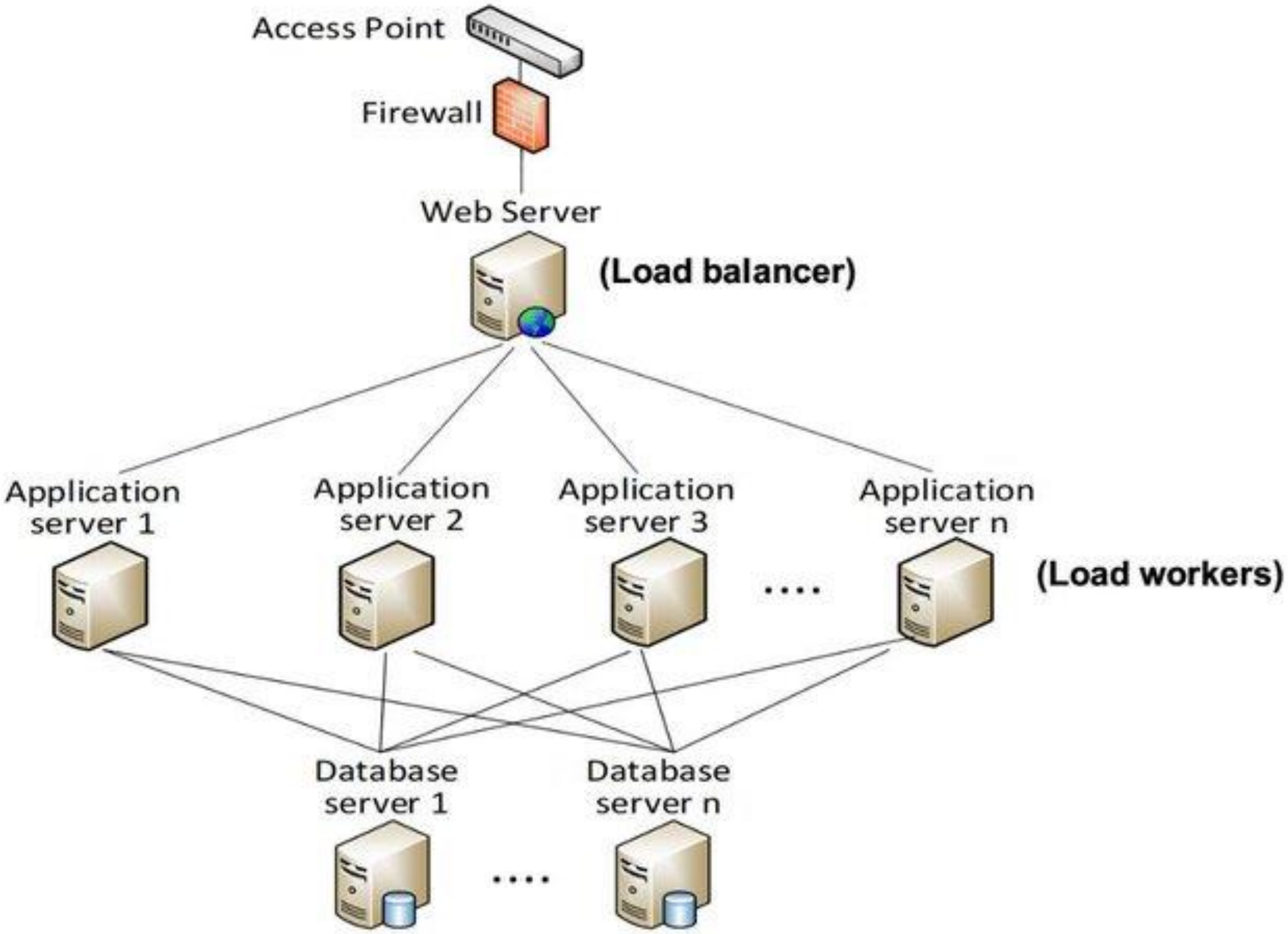
1. Network Routing
   - This involves how data packets are directed from the server to other network devices.

2. Application-Level Routing
   - This refers to how incoming requests are handled within the server itself.

Routing

Application-Level Routing

- Load Balancing
  - For servers handling high traffic, load balancing distributes incoming requests across multiple servers to improve performance and reliability.
- Content Delivery Networks (CDNs)
  - These networks optimize content delivery by caching static content closer to end users.
- Application-Specific Routing
  - Some applications may have their own routing mechanisms, such as routing incoming HTTP requests to specific handlers or controllers.


Access Point
Firewall
Web Server
(Load balancer)
Application server 1
Application server 2
Application server 3
Application server n
(Load workers)
Database server 1
Database server n

Transport layer

Transport layer

- The transport layer can **maintain flow control**, and can provide **error checking** and **recovery of data** between network devices

- The most used transport layer protocols are **TCP** and **UDP**

TCP

- Transmission Control Protocol (TCP) uses the IP protocol to create **reliable transmission** of so-called TCP/IP packets

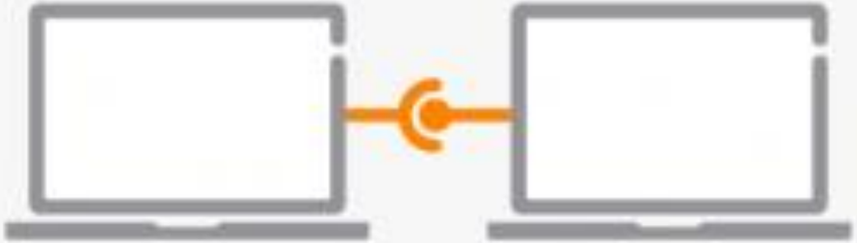
- TCP provides **reliable**, **ordered delivery** of a stream of data between applications

- TCP introduces much **overhead**

UDP

- User Datagram Protocol (UDP) emphasizes **reduced latency** over reliability
- It sends data without checking if the data arrived
  - Reduces much overhead
- UDP is typically used when **some packet loss is acceptable**
  - Real-time voice and video streams
  - When only small amounts of data are transmitted, that fit in one IP packet

TCP



- Slower but more reliable transfers
- Typical Applications:
  - File Transfer Protocol (FTP)
  - Web Browsing
  - Email

unicast

UDP

- Faster but not guaranteed transfers ("best effort")
- Typical Applications:
  - Live Streaming
  - Online Games
  - VoIP

unicast multicast broadcast

TCP and UDP ports

- TCP and UDP use logical port numbers
- Each side of a TCP or UDP connection uses an associated port number between 0 and 65,535
- Received TCP or UDP packets are identified as belonging to a specific connection by its combination of the IP address, and the TCP or UDP port number
  - For instance: 192.168.1.2:80, the number after the colon represents the port number (80 in this case)

TCP and UDP ports

- Servers running a specific service listen to well-known ports:
  - FTP (port 21
  - SSH (port 22)
  - SMTP (port 25)
  - DNS (port 53)
  - HTTP (port 80) and HPPTS (port 443)

DNS - Domain Name System

- Port 53 is specifically for DNS. Other services should not use this port.
- DNS queries are typically small, so UDP is efficient.
- For larger DNS responses or when reliability is critical, TCP is used.
- Firewalls should allow traffic on port 53 for DNS to function correctly.

DNS

Domain Name	=	IP Address
google.com	⟷	173.194.77.104
twitter.com	⟷	199.59.149.230
whitehouse.org	⟷	67.132.30.211
bluemoonstudio.org	⟷	97.74.144.192
facebook.com	⟷	173.252.110.27

Database Servers

Database	Default Port
MySQL	3306
PostgreSQL	5432
Microsoft SQL Server	1433
Oracle Database	1521
MongoDB	27017

Session layer

Session layer

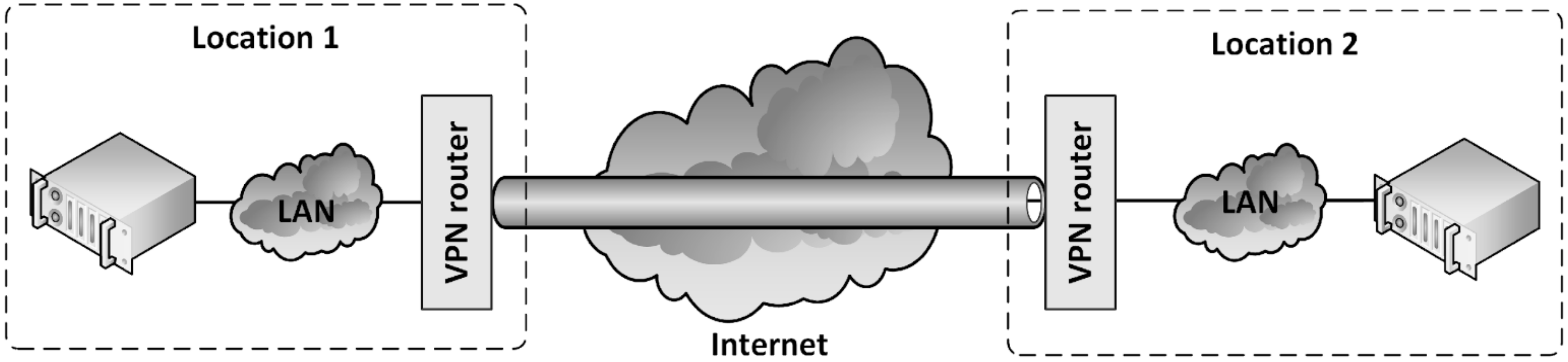
- The session layer provides mechanisms for opening, closing and managing a session between end-user application processes

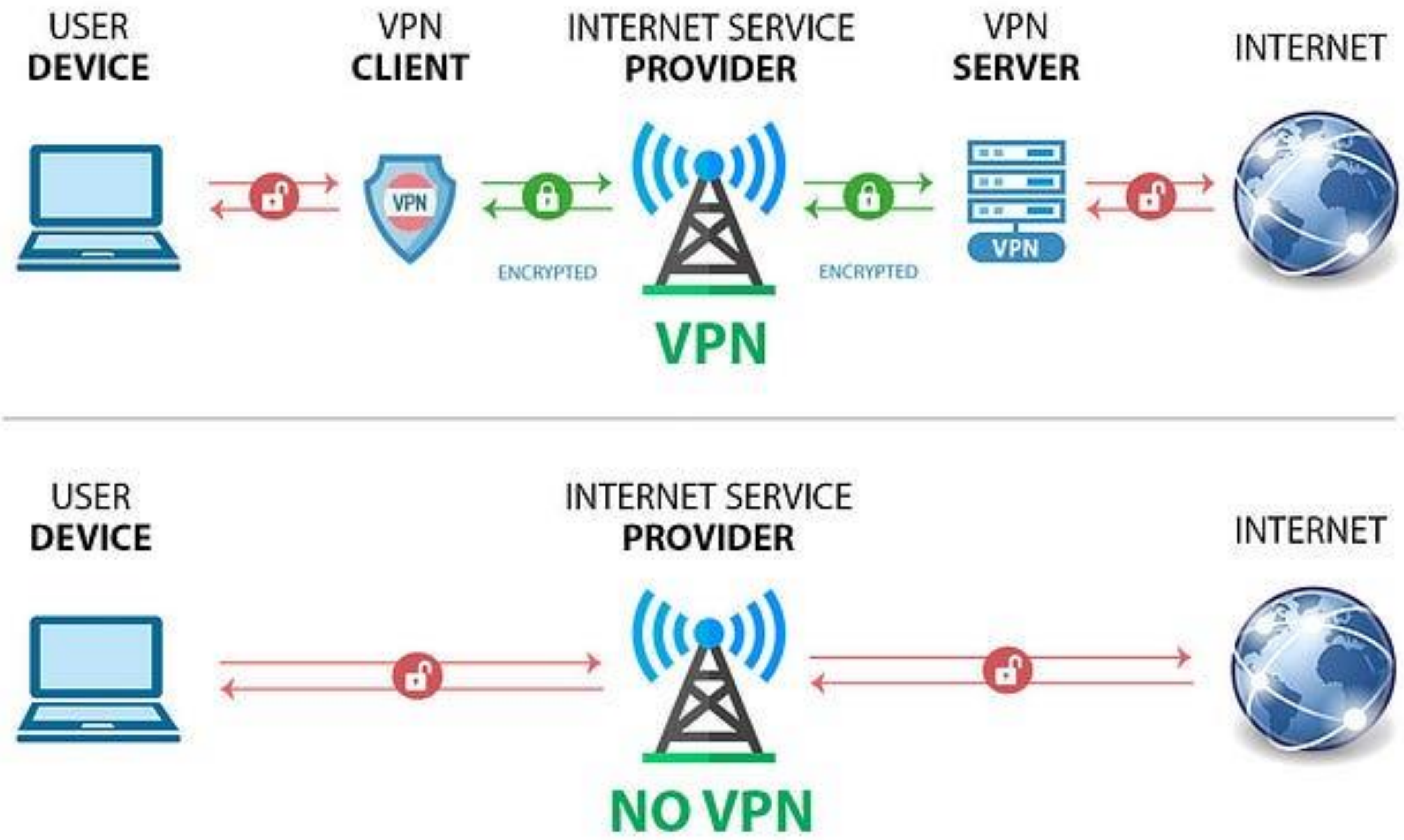
Virtual Private Network (VPN)

- A Virtual Private Network (VPN) uses a public network to interconnect private sites in a secure way. Also known as a VPN tunnel

- VPN uses "virtual" connections based on IPsec/SSL

- Most network providers also offer private VPNs based on MPLS




USER DEVICE
VPN CLIENT
INTERNET SERVICE PROVIDER
VPN SERVER
INTERNET
VPN
ENCRYPTED
ENCRYPTED
VPN
VPN
USER DEVICE
INTERNET SERVICE PROVIDER
INTERNET
NO VPN

Virtual Private Network (VPN)

- VPNs use **strong encryption** and **strong user authentication**. Using the internet for transmitting sensitive data is considered safe
- VPN tunnels are often used for remote access to the LAN by users outside of the organization's premises
- Most common VPN communications protocol standards:
  - Point-to-Point Tunneling Protocol (PPTP) for individual client to server connections
  - Layer 2 Tunneling Protocol (L2TP) for individual client to server connections
  - IPsec for network-to-network connectivity
- IPsec is built into IPv6 standard and is implemented as an add-on to IPv4

Presentation layer

Presentation layer

- This **layer takes the data** provided by the application layer and **converts it into a standard format** that the other layers can **understand**

- Many protocols are implemented in the presentation layer.

- Secure Sockets Layer (SSL) and Transport Layer Security (TLS) are the most important ones.

SSL and TLS

- Allow applications to communicate securely over the internet using data encryption
- Secure Sockets Layer (SSL)
  - SSL is considered insecure and should not be used
- Transport Layer Security (TLS)
  - TLS is securing WWW traffic carried by HTTP to form HTTPS
  - TLS version 1.2 is still secure
  - TLS 1.3 offers a faster TLS handshake and simpler, more secure cipher suites
  - TLS relies on an application capable of handling the protocol (like a Web browser)

Application layer

Application layer

- The topmost layer where servers interact directly with user applications.

- This layer interacts with the operating system or application

- This layer also contains the relatively simple infrastructure services

- These infrastructure services are used by the infrastructure itself
  - Not necessarily used by upper layer applications

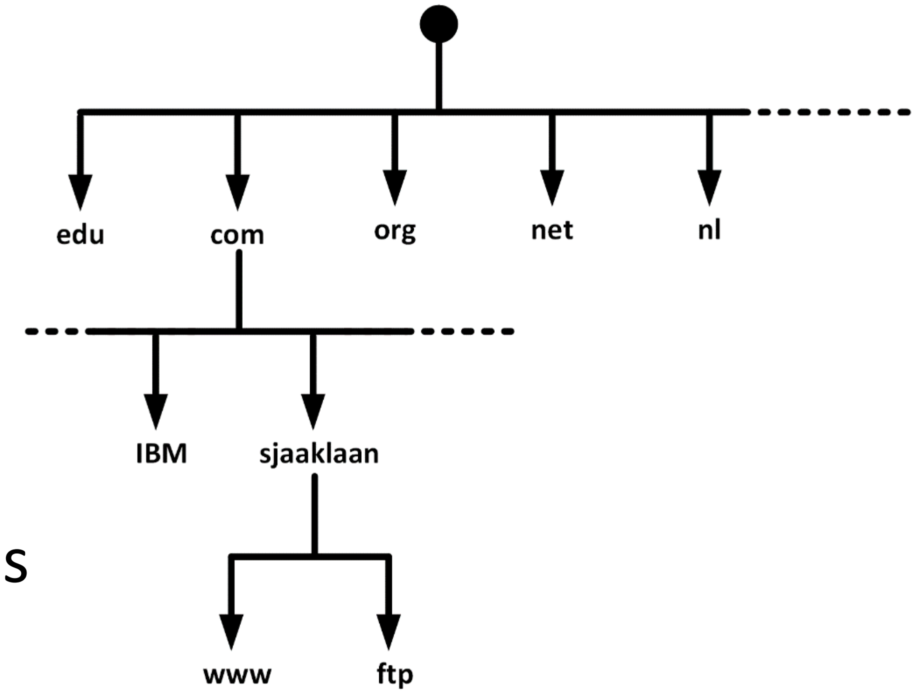
- If infrastructure services fail, usually the entire infrastructure fails!

Role of the Application Layer in Servers

- **Protocol Implementation**: Servers implement various application layer protocols to provide specific services. For example, a web server implements HTTP, an email server uses SMTP, POP3, or IMAP, and a file transfer server uses FTP.
- **Data Formatting**: The application layer ensures data is formatted correctly for the specific application and the underlying network.
- **Error Handling**: It handles errors that occur during data transmission and provides mechanisms for retransmission or recovery.
- **Security**: The application layer can implement security measures like encryption and authentication.
- **Session Management**: It manages communication sessions between clients and servers.

DNS

- For example, www.tarc.edu.my is translated to 103.52.192.135
- This IP address is used by the browser to connect to the web server
- DNS distributes the responsibility of mapping domain names to IP addresses by designating authoritative name servers for each domain
- DNS is a distributed database that links IP addresses with domain names
- Translates domain names, meaningful to humans, into IP addresses



DNSSEC

- DNS has a number of security issues
  - DNS was not designed with security in mind
  - Updates to DNS records are done in non-encrypted clear text
  - Authorization is based on IP addresses only
- DNSSEC is a set of extensions to DNS
  - Provides origin authentication of DNS data
  - Provides data integrity
- DNSSEC is not in wide spread use today
  - All DNS servers must implement DNSSEC in order to make full use of all benefits

IPAM systems

- IP address management (IPAM) systems are appliances that can be used to plan, track, and manage IP addresses in a network

- IPAM systems integrate DNS, DHCP, and IP address administration in one high available redundant set of appliances

Network Time Protocol (NTP)

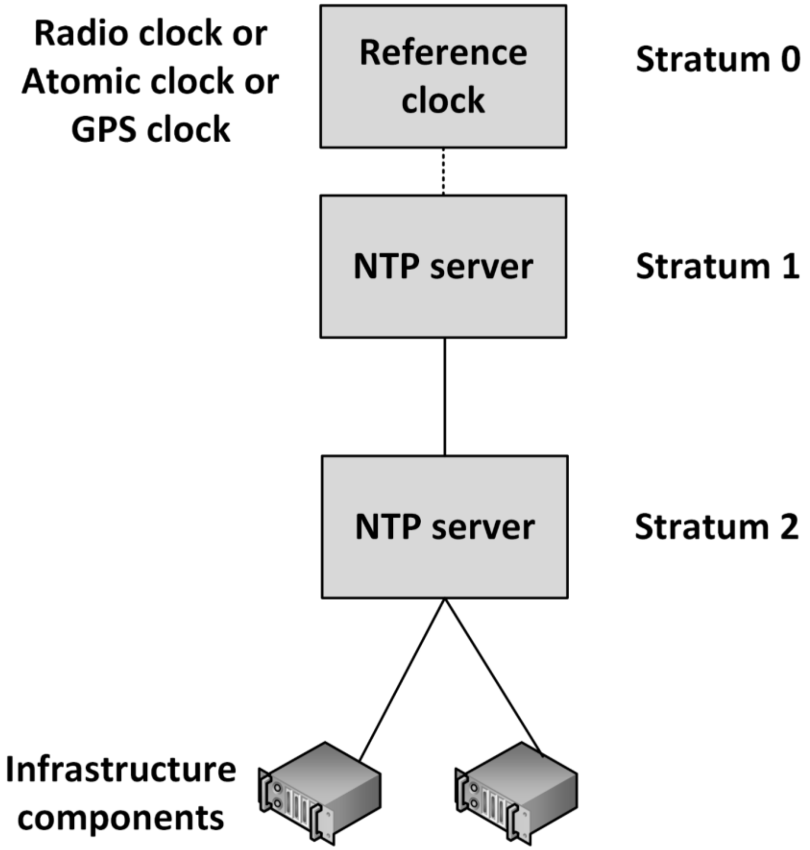
- NTP ensures all infrastructure components use the same time in their real-time clocks
- Particularly important for:
    - Log file analysis
    - Clustering software
    - Kerberos authentication
- NTP can maintain time:
    - To within 10 milliseconds over the internet
    - Accurate to 0.2 milliseconds or better in LANs
- When the time in an operating system is incorrect, the NTP client in the operating system changes the operating system clock

Network Time Protocol (NTP)

- NTP servers can be implemented as:
  - Software on operating systems, routers, and switches
  - Dedicated hardware appliances – often using some external signal like long wave radio clocks or GPS clocks
  - NTP time synchronization services on the internet
- NTP provides time in Coordinated Universal Time (UTC, previously known as GMT)
- The translation to the local time zone, including the switch to and from daylight saving time, is done at the operating system level, not in NTP clocks

Network Time Protocol (NTP)

- NTP operates within a hierarchy
- Each level in the hierarchy is assigned a number called the stratum
- The stratum defines its distance from the reference clock



POP and SMTP

- POP = Post Office Protocol. Used by email clients to retrieve email messages from a mail server
  - POP does not synchronize email messages between the client and server. If a user reads or deletes an email message on their email client, it will not be synchronized back to the mail server
- SMTP = Simple Mail Transport Protocol. Used to send email messages from a mail client to a mail server or between mail servers
  - When a user sends an email message, it is sent to the user's mail server, which then forwards the message to the recipient's mail server using the SMTP protocol. The recipient's mail server then delivers the message to the recipient's inbox
  - SMTP is a text-based protocol. Email messages are transmitted in plain text between the client and the server
  - SMTP supports TLS encryption, to create secure email communications
- MIME = Multipurpose Internet Mail Extensions.
  - Enables SMTP to support file attachments in email messages
  - MIME converts an attachment into a stream of ASCII characters and then embeds those characters in the e-mail message

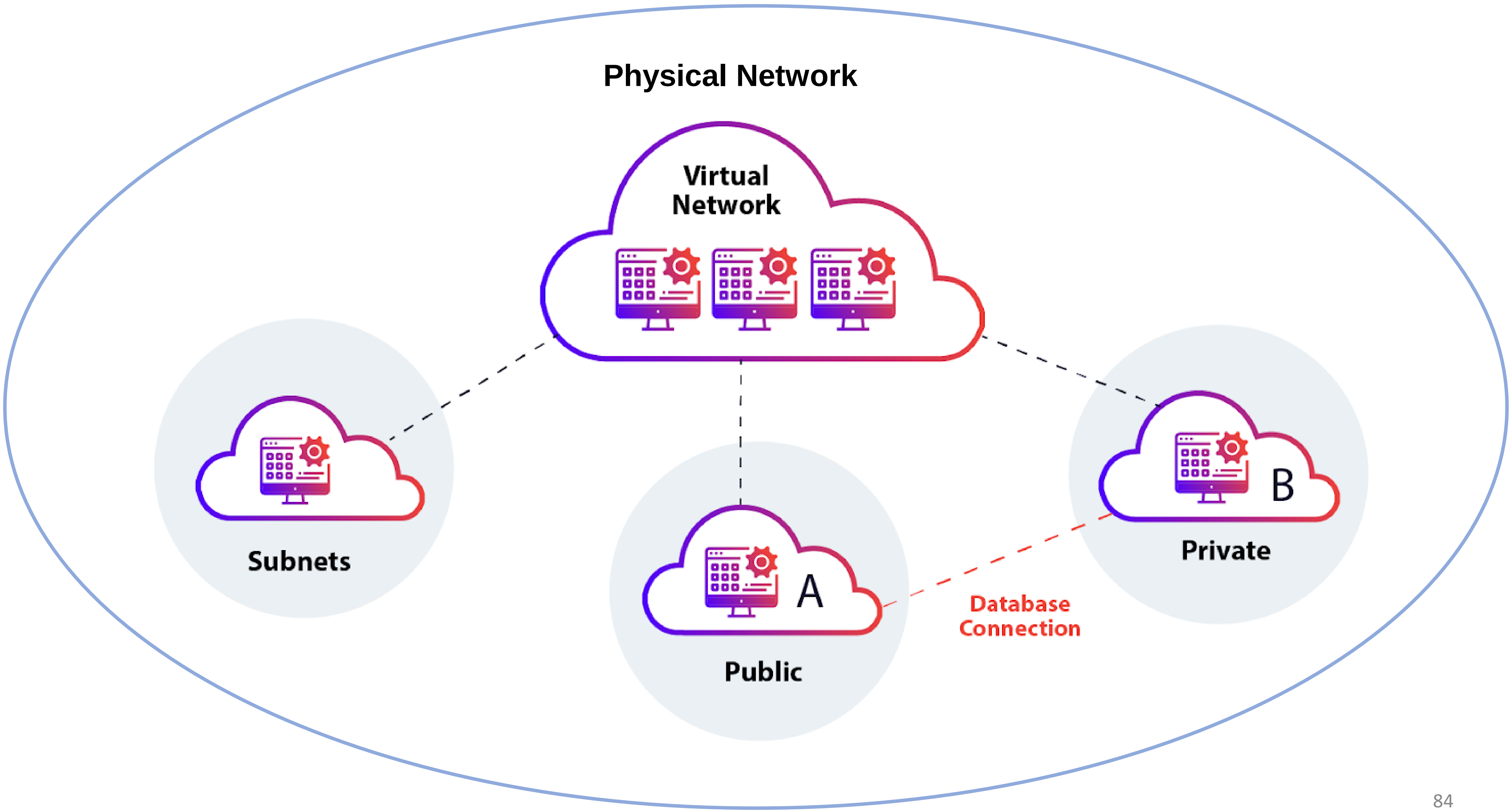
FTP

- FTP = File Transfer Protocol. A protocol for transferring files between computers
  - The FTP client software initiates a connection to the server and sends commands to request and transfer files
  - FTP includes features such as file listing, directory creation, and file deletion
- FTP can operate in active or passive mode
  - In active mode, the client opens a TCP port and the server establishes a connection to the client
  - In passive mode, the server opens a TCP port (often the default port 21) and the client connects to the server. Passive mode is often used to get around firewall restrictions that prevent incoming connections
- FTP has some security concerns, such as sending login credentials over the network in clear text
  - FTPS (FTP Secure) and SFTP (Secure File Transfer Protocol) are alternatives to FTP that provide encryption of data in transit

HTTP and HTTPS

- HTTP = Hypertext Transfer Protocol
- HTTPS = HTTP Secure
- Both are protocols to transfer data over the internet
  - HTTP(S) is most commonly used when browsing the web with a web browser such as Chrome, Firefox, Safari, or Edge
- HTTP defines how messages are formatted and transmitted, and how web servers and browsers should respond to various commands
  - When a user types a web address into a web browser, the browser sends an HTTP request to the web server, which responds with an HTTP response containing the requested resource, such as a web page or image.
- HTTPS is a secure version of HTTP
  - It adds an extra layer of security using TLS

Network virtualization


Physical Network
Virtual
Network
Subnets
A
Public
B
Private
Database
Connection

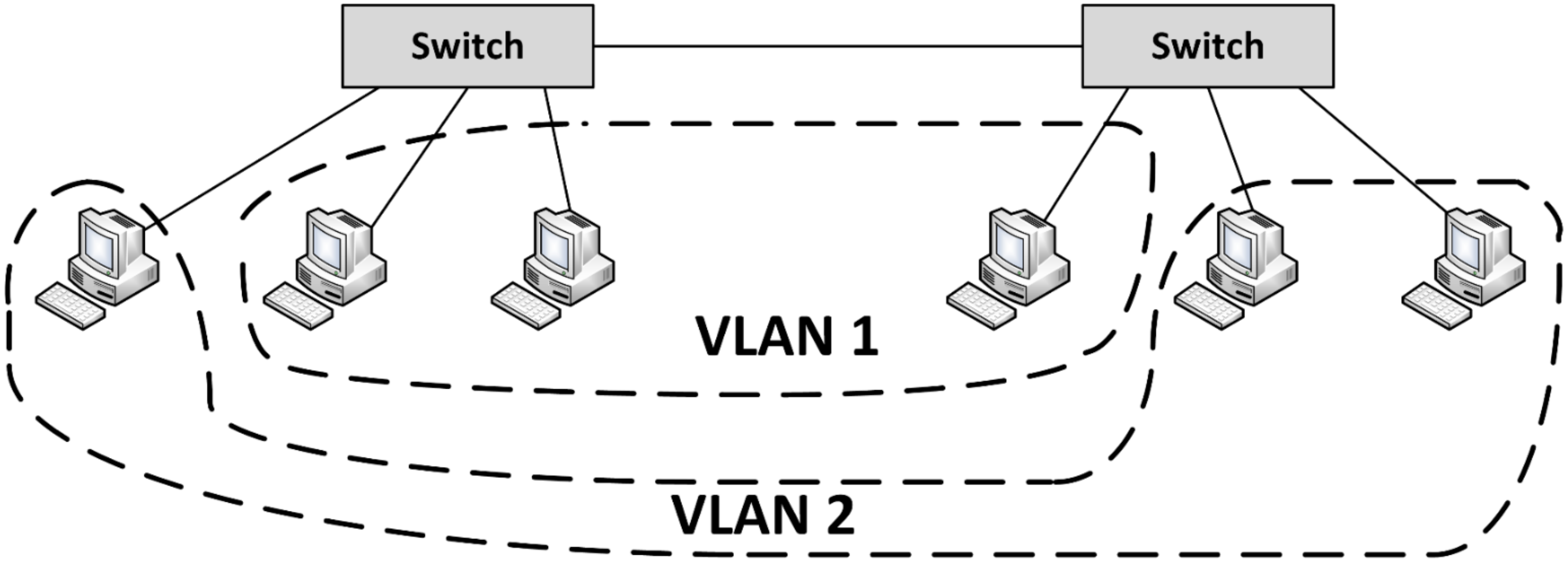
Network virtualization

Common technologies used for network virtualization:

1. Virtual Local Area Network (VLAN)
2. Virtual Extensible LAN (VXLAN)
3. Software-Defined Networking (SDN)

Virtual LAN (VLAN)

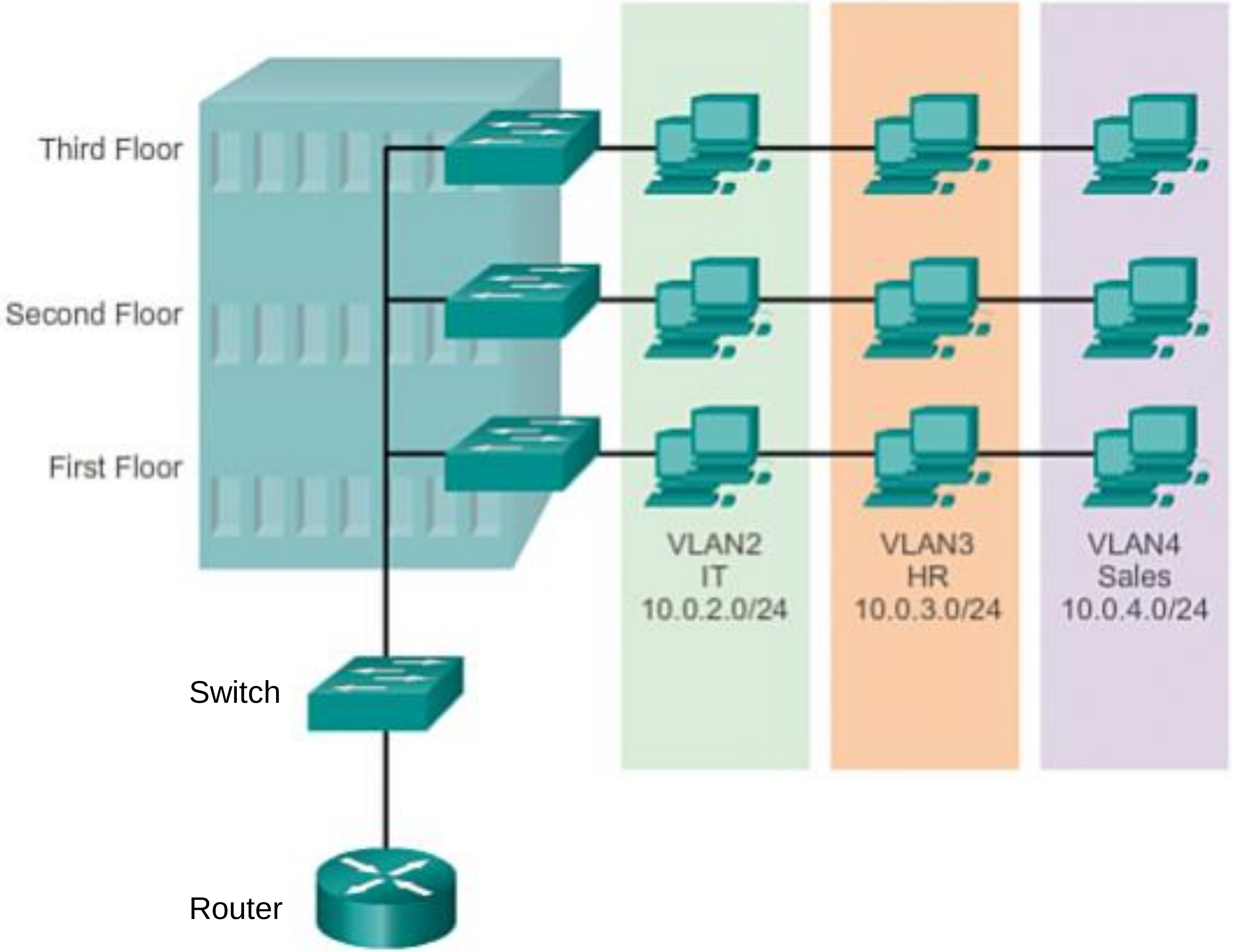
- VLANs enable logical grouping of network nodes on the same LAN
  - Configured on network switches
  - Operate at the Ethernet level



Virtual LAN (VLAN)

- VLANs:
  - Allow **segmenting a network** at the data link layer
  - Allow end stations to be **grouped together** even if they are **not physically connected** to the same switch
  - Can adapt to changes in network requirements and allow **simplified administration**
  - **Enhance security** by preventing traffic in one VLAN from being seen by hosts in a different VLAN

- For VLANs to **communicate** with each other a **router** is needed

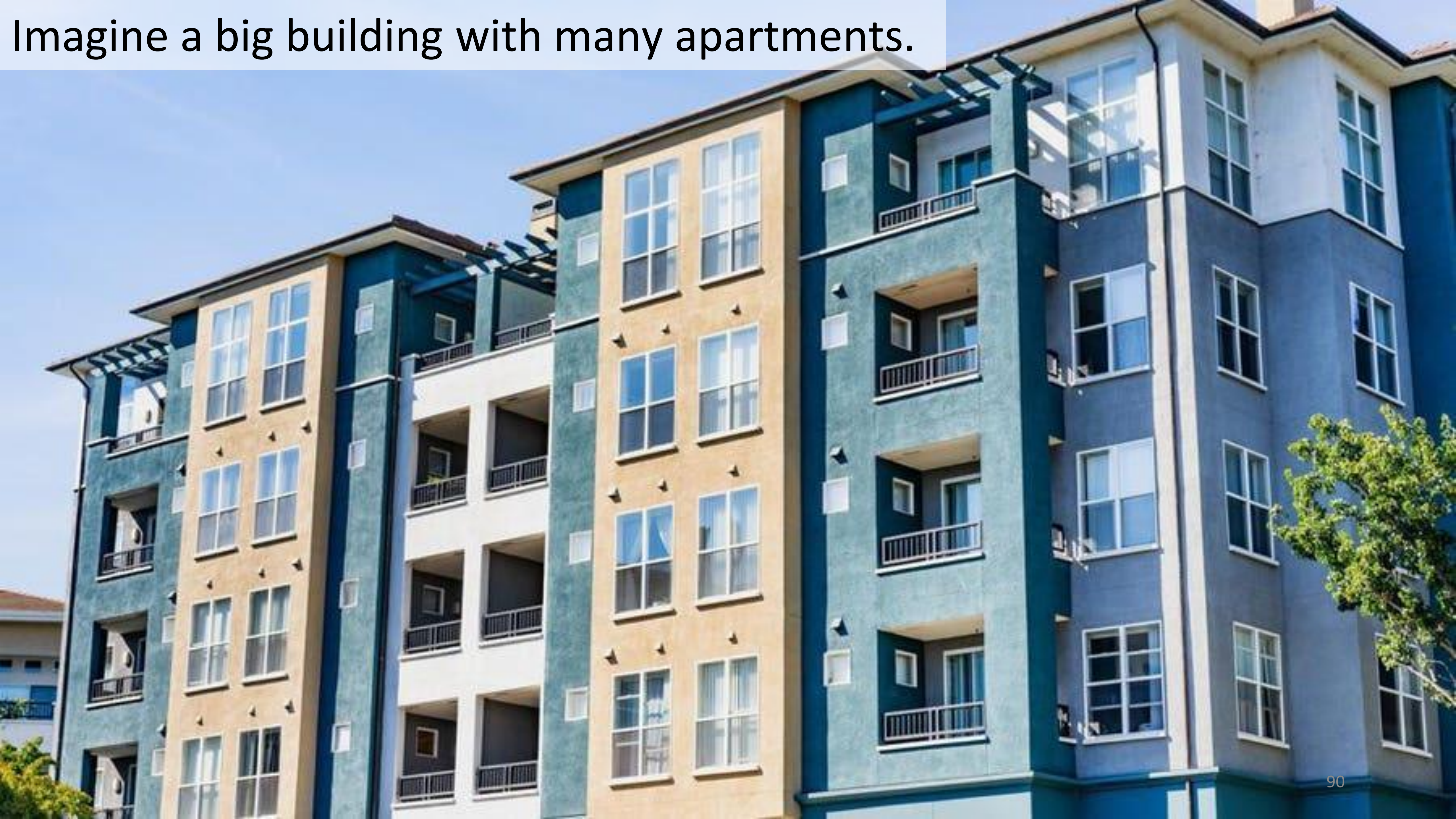


Create separate VLANs for each department to isolate network traffic and improve security. For instance, the IT department might have VLAN2, HR VLAN3, and Sales VLAN4.

Virtual Extensible LAN (VXLAN)

- VXLAN is an **encapsulation protocol**
- Can be used to create a logical switched layer 2 network across routed layer 3 networks
- Only servers **within the same logical network can communicate** with each other
- VXLANs are heavily used in **multi-tenant cloud environments**

Imagine a big building with many apartments.



Each apartment is like a separate VLAN

VXLAN allows you to create many virtual apartments (VLANs) within the same building

Virtual Routing and Forwarding (VRF)

- It allows **multiple** instances of a **routing table** to coexist within the same router
  - This allows **multiple virtual IP networks** to be created within a **single physical network** infrastructure
  - VRF provides features similar to VLANs, but at **layer 3** of the OSI model instead of layer 2

Virtual Routing and Forwarding (VRF)

- A router that supports VRF can have **multiple virtual routers** implemented.
  - One or more interfaces on the router can be part of a VRF, but none of the VRFs share routing information. Packets are only forwarded between interfaces that are in the same VRF

- **Overlapping IP addresses** can be used because the different routing instances are independent of each other

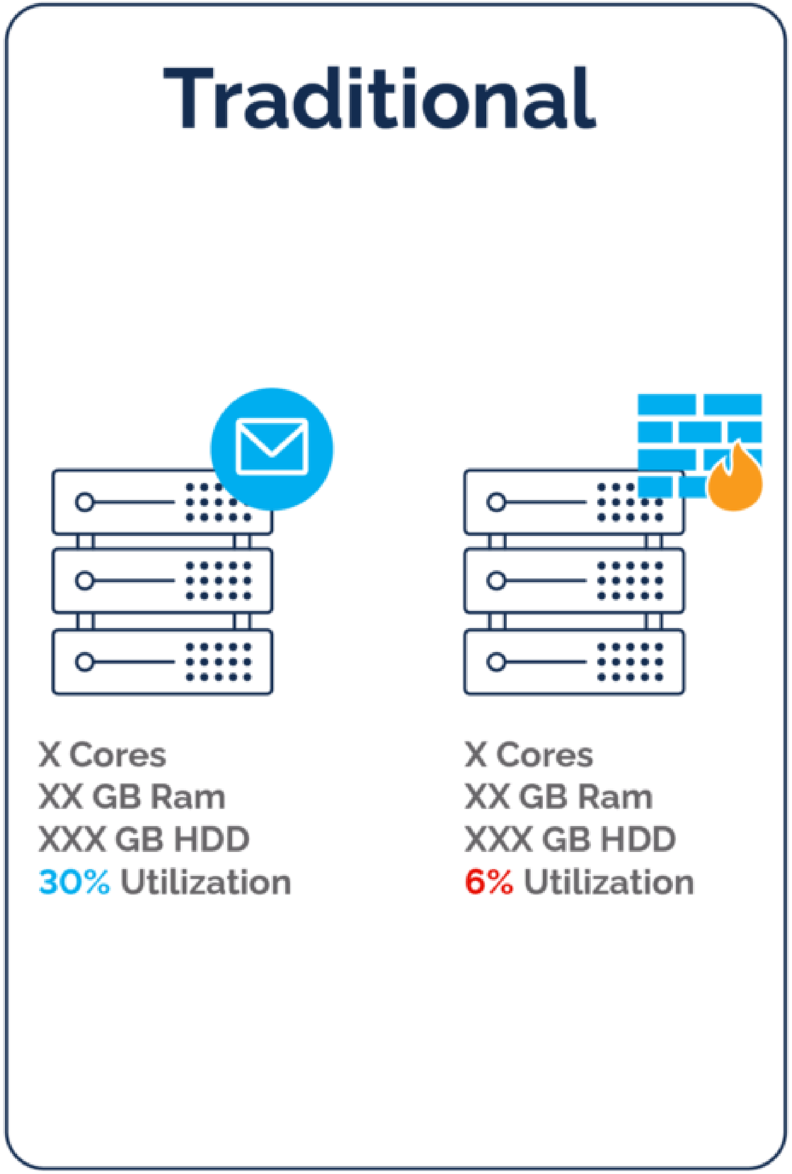
Virtual NICs

- **Virtual machines** are only aware of **virtual Network Interface Controllers** (NICs) provided to them

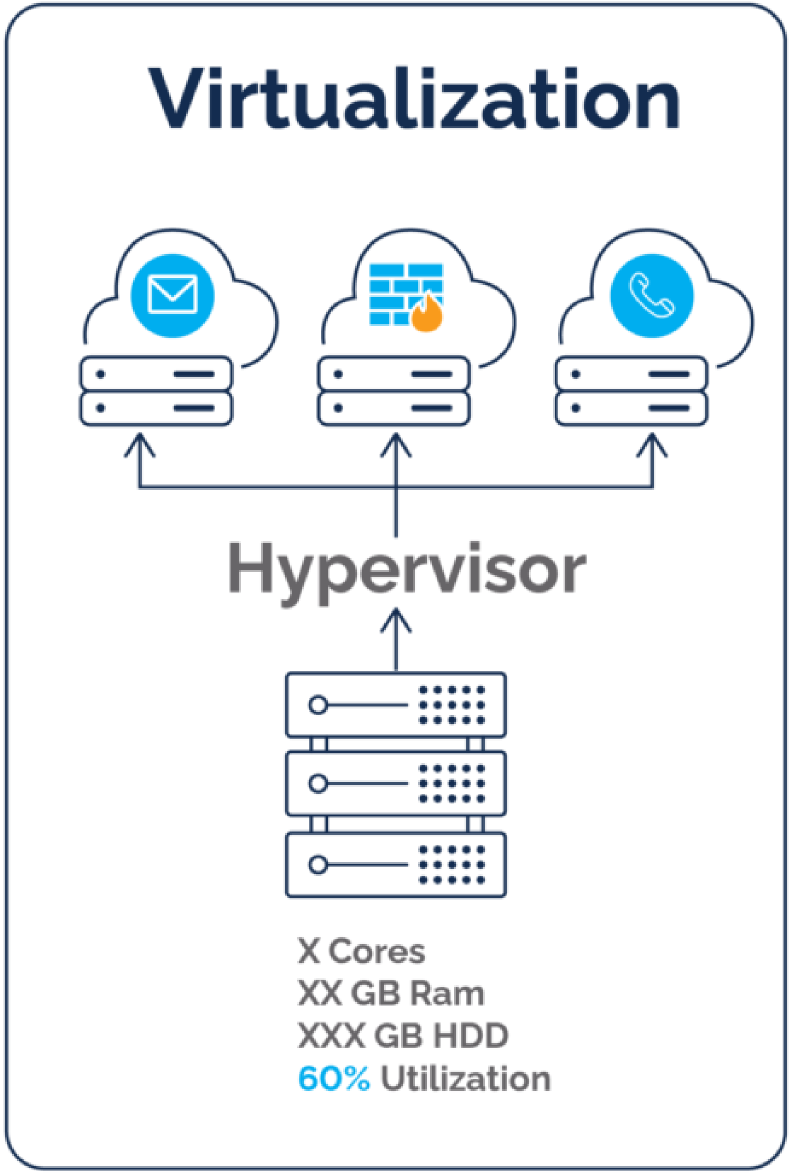
- **Virtual machines** running on physical machines **share physical NICs**

- **Communications between virtual machines** on the same physical machine are routed directly **in memory** space by the **hypervisor**, without using the physical NIC

- The hypervisor routes Ethernet packages from the virtual NIC on the virtual machine to the physical NIC on the physical machine



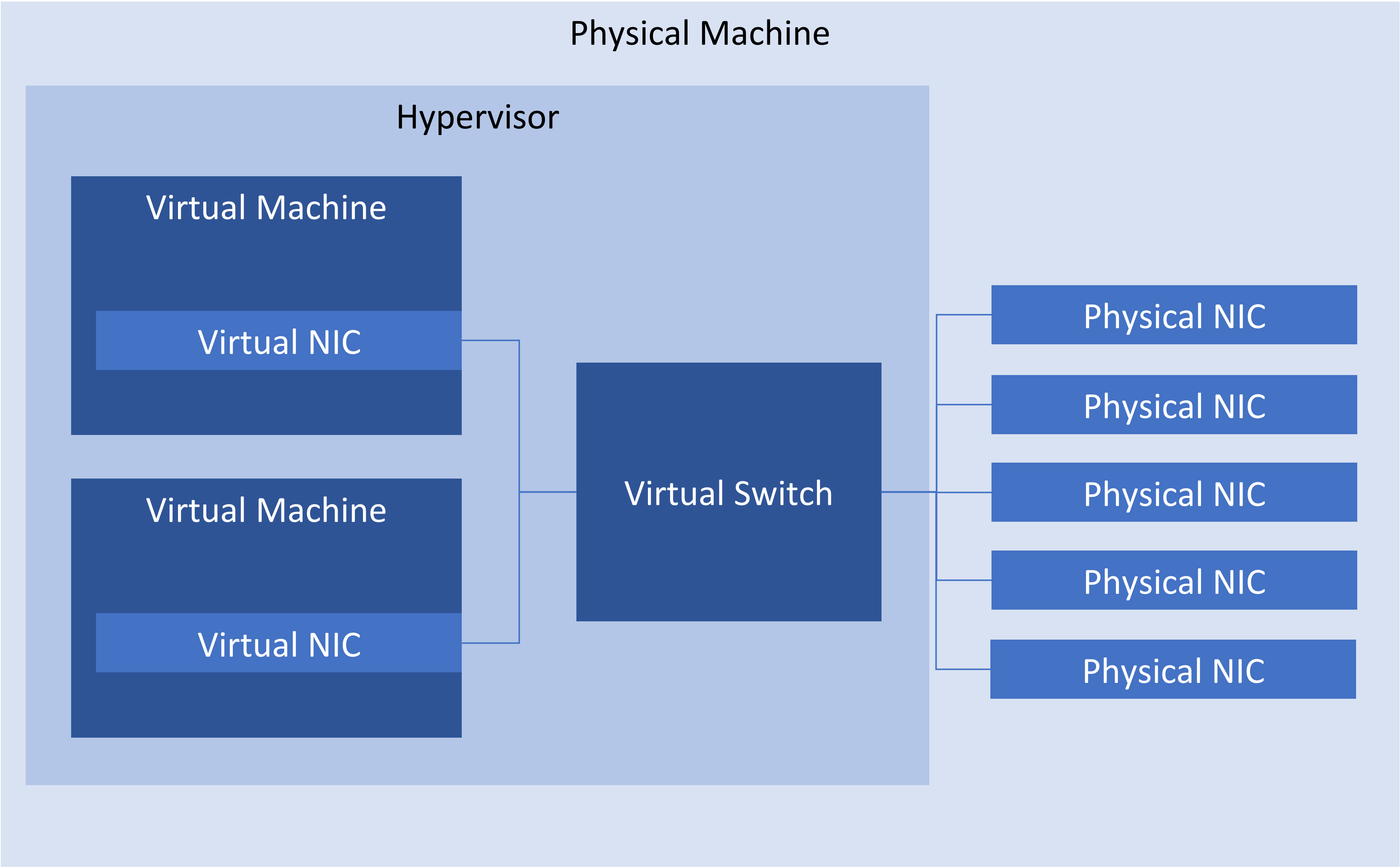
Traditional
X Cores
XX GB Ram
XXX GB HDD
30% Utilization
X Cores
XX GB Ram
XXX GB HDD
6% Utilization



Virtualization
Hypervisor
X Cores
XX GB Ram
XXX GB HDD
60% Utilization

Virtual switch

- **Virtual NICs** are connected to **virtual switches**
- A virtual switch is an application running in the **hypervisor**, with most of the capabilities of a physical network switch
- A virtual switch is dynamically configured
  - Ports in the virtual switch are configured at runtime
  - The number of ports on the switch is in theory unlimited


Physical Machine
Hypervisor
Virtual Machine
Virtual NIC
Virtual Machine
Virtual NIC
Virtual Switch
Physical NIC
Physical NIC
Physical NIC
Physical NIC
Physical NIC

Virtual switch

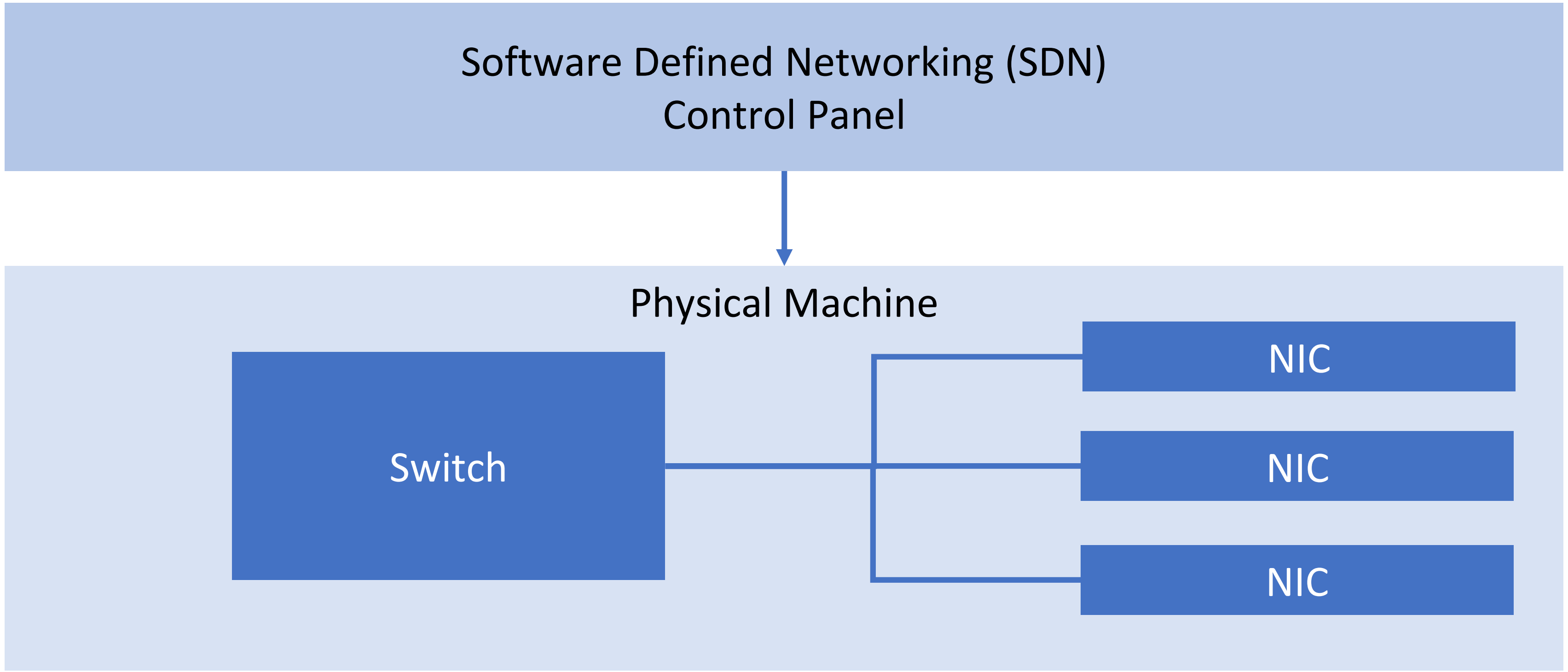
- Availability:
  - No cable disconnects
  - No need for auto-detecting network speed
  - No network hubs, routers, adapters, or cables that could physically fail
- Security:
  - No easy way to intercept network communications between virtual machines from outside of the physical machine

Software Defined Networking

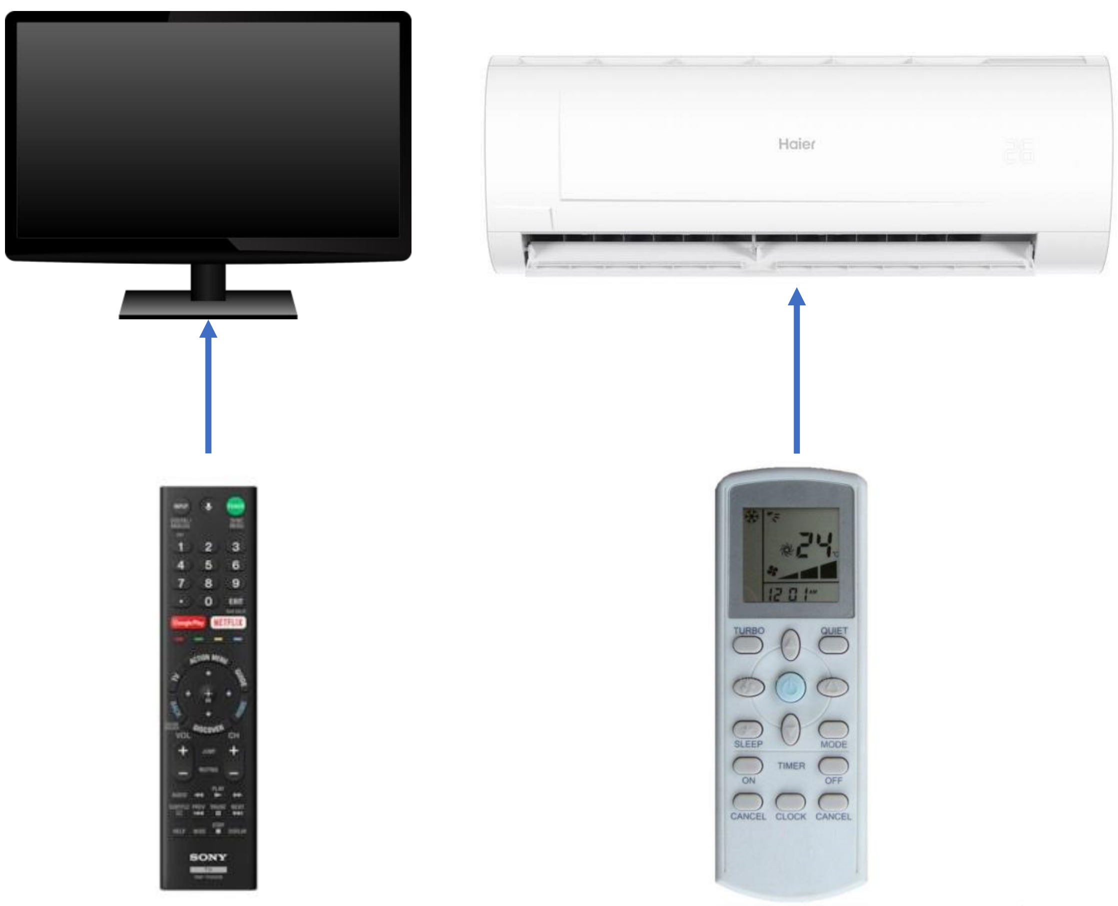
- Software Defined Networking (SDN) allows networks to be defined and controlled using software external to the physical networking devices

- A set of physical network switches can be programmed as a virtual network:
  - Hierarchical
  - Complex
  - Secured

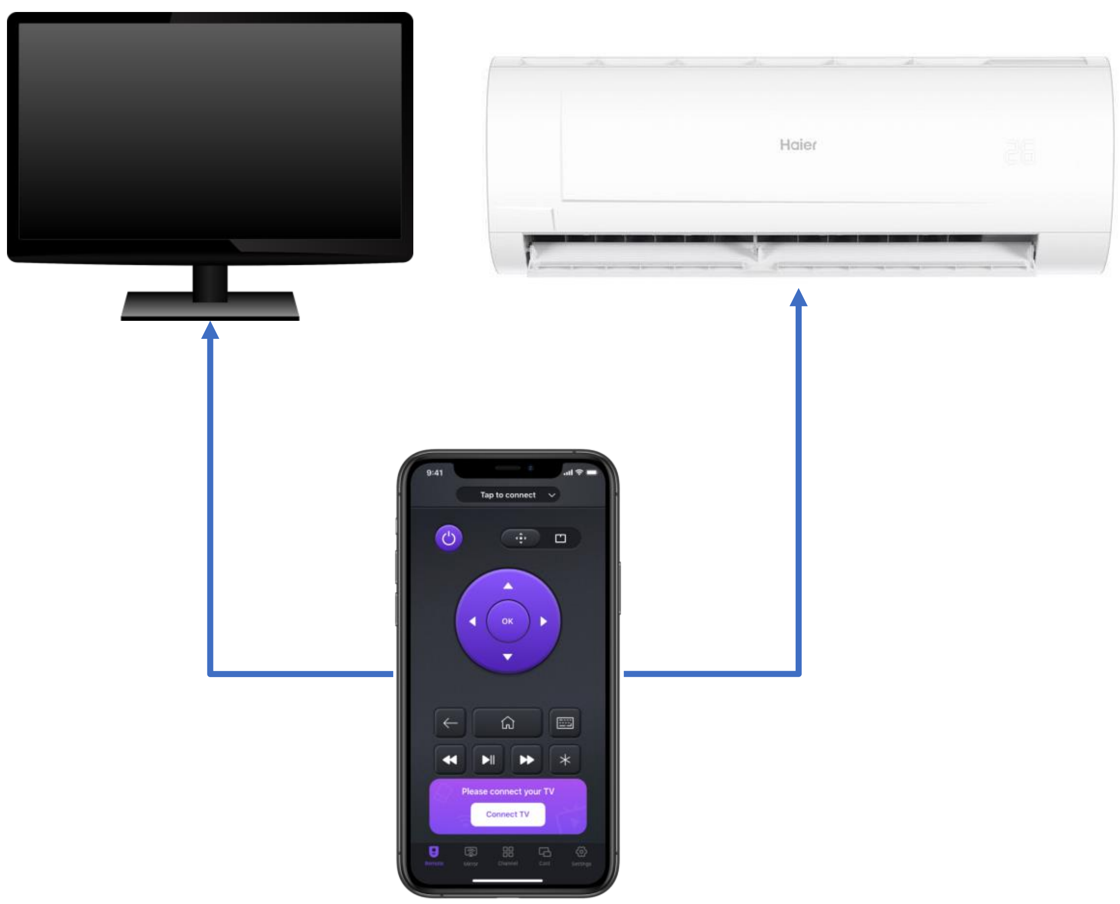
- A virtual network can easily be changed without touching the physical network components


Software Defined Networking (SDN)
Control Panel
Physical Machine
Switch
NIC
NIC
NIC

Physical



Virtual



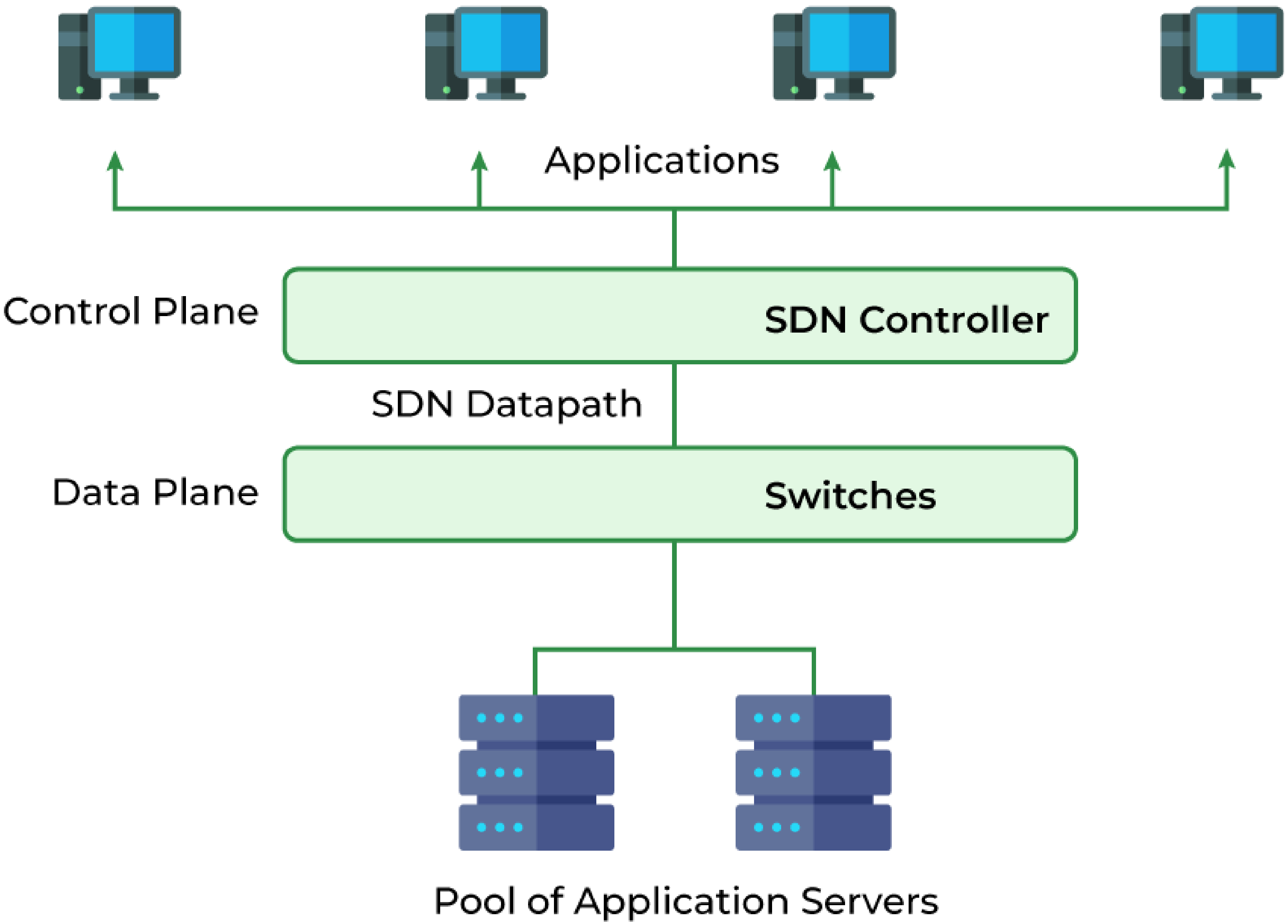
Software Defined Networking

- It **separates the control plane** (deciding how to route traffic) **from the data plane** (actually forwarding traffic) in network devices.

- Control plane resides centrally

- Data plane (the physical switches) remain distributed

Software Defined Networking (SDN)



Software Defined Networking

- SDN can be controlled from a single management console
- Provides open APIs that can be used to manage the network using third party software
- In an SDN, the distributed data plane devices are only forwarding network packets based on Address Resolution Protocol (ARP) or routing rules that are preloaded into the devices by the SDN controller in the control plane. This allows the physical devices to be much simpler and more cost effective

Software Defined Networking

- Good examples of SDN are implemented by the public cloud providers
  - AWS has Virtual Private Clouds (VPCs)
  - Azure has Virtual Networks (VNETs)
  - GCP has Virtual Private Cloud

Network Function Virtualization

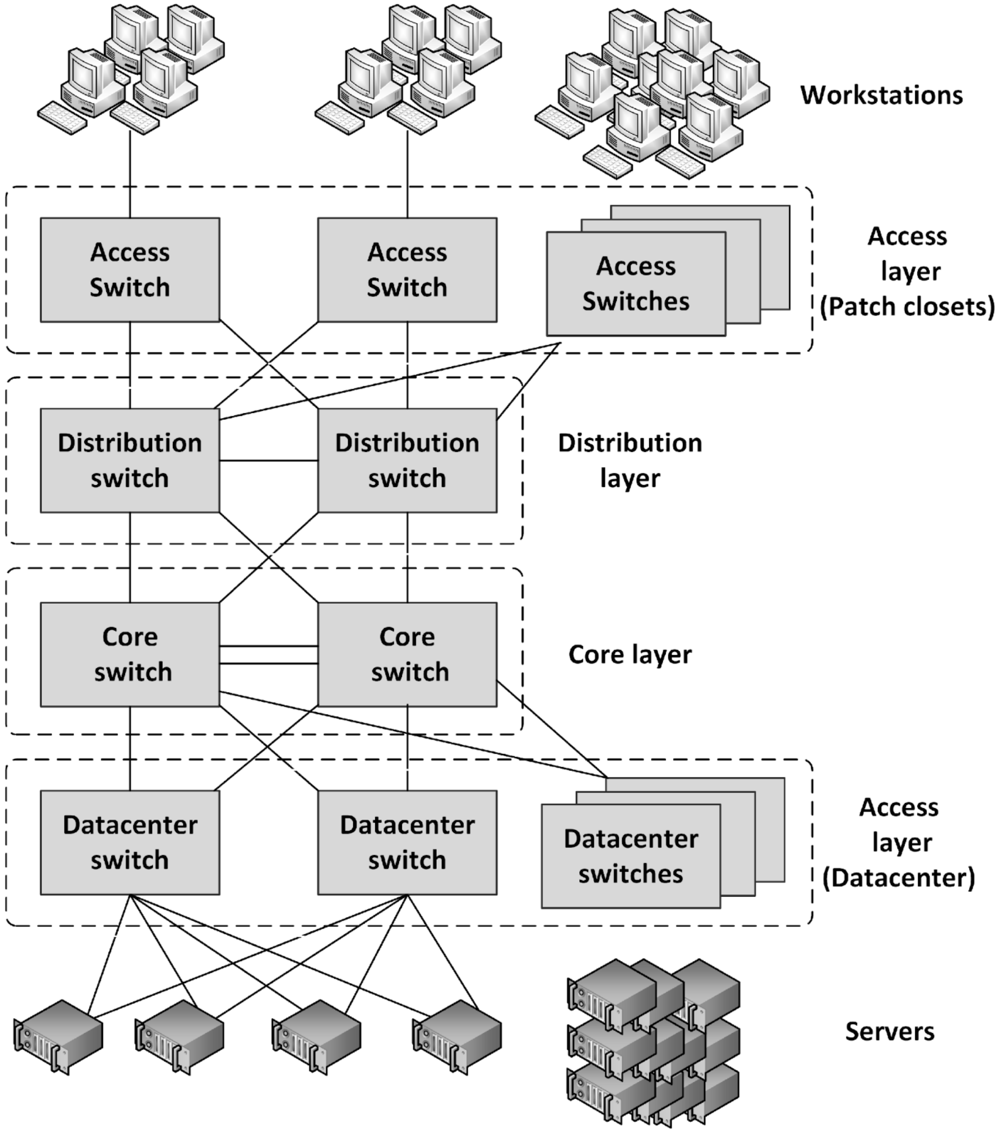
- Network Function Virtualization (NFV) is a way to **virtualize networking devices**. Examples, firewalls, VPN gateways, and load balancers
- NFV appliances **are implemented as virtual machines** running applications that perform the network functions
- NFV virtual appliances can be **created** and **configured** dynamically and on-demand using **APIs**. Example: Deploy a new firewall as part of a script that creates a number of connected virtual machines in a cloud environment

A REST API call to create a new virtual network on an SDN controller.


```json
{
  "method": "POST",
  "url": "/v1/networks",
  "data": {
    "name": "my-network",
    "subnet": "192.168.1.0/24"
  }
}
```

Network availability

Layered network topology



Layered network topology

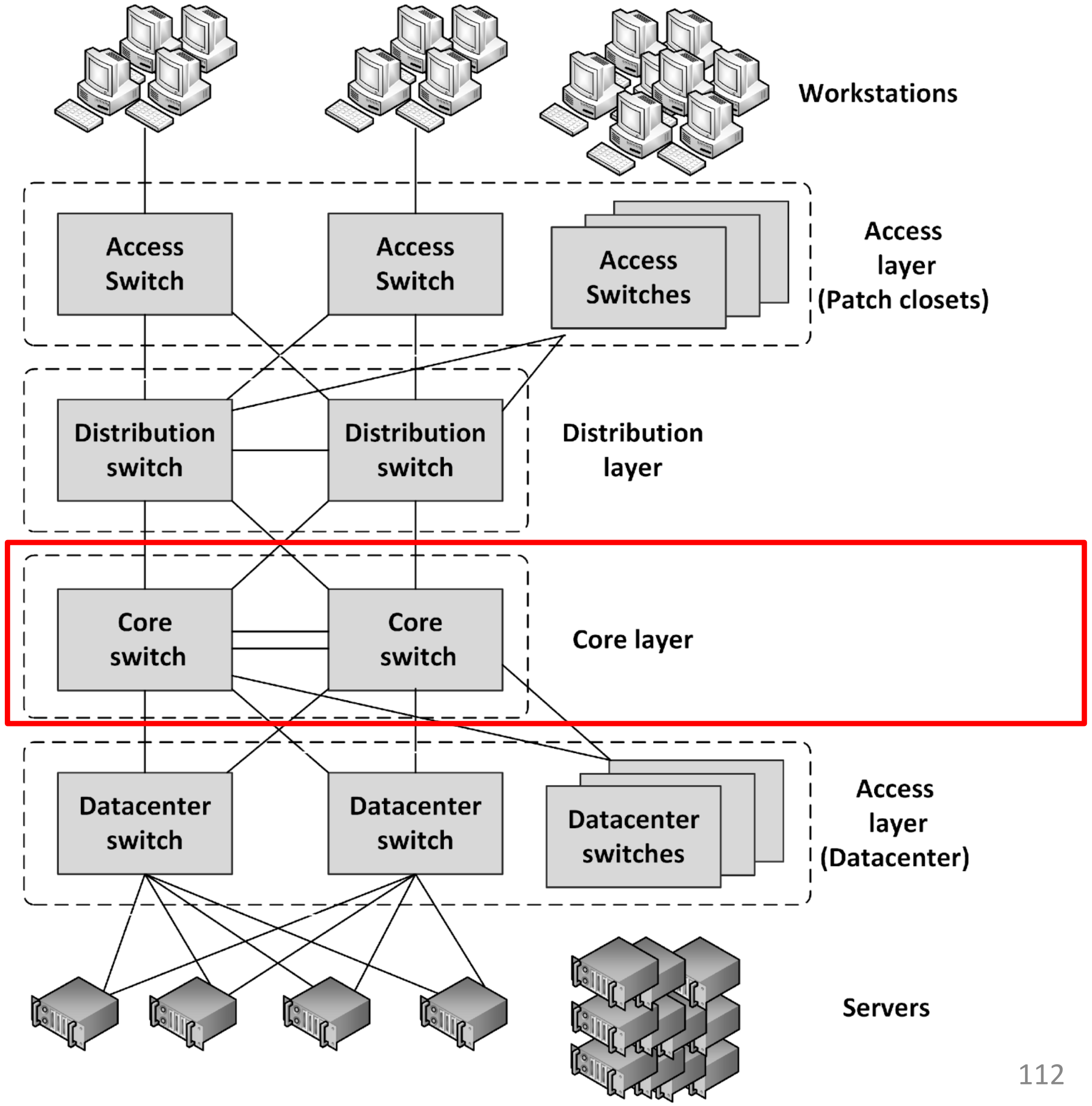
- A network infrastructure should be built up in layers
  - Improve availability and performance
  - Provides scalability
  - Provides deterministic routing : advance determination of the routes between given pairs of nodes
  - Avoids unmanaged ad-hoc data streams

- Provides high availability because the layering **provides multiple paths** to any piece of equipment

Layered network topology

Core layer

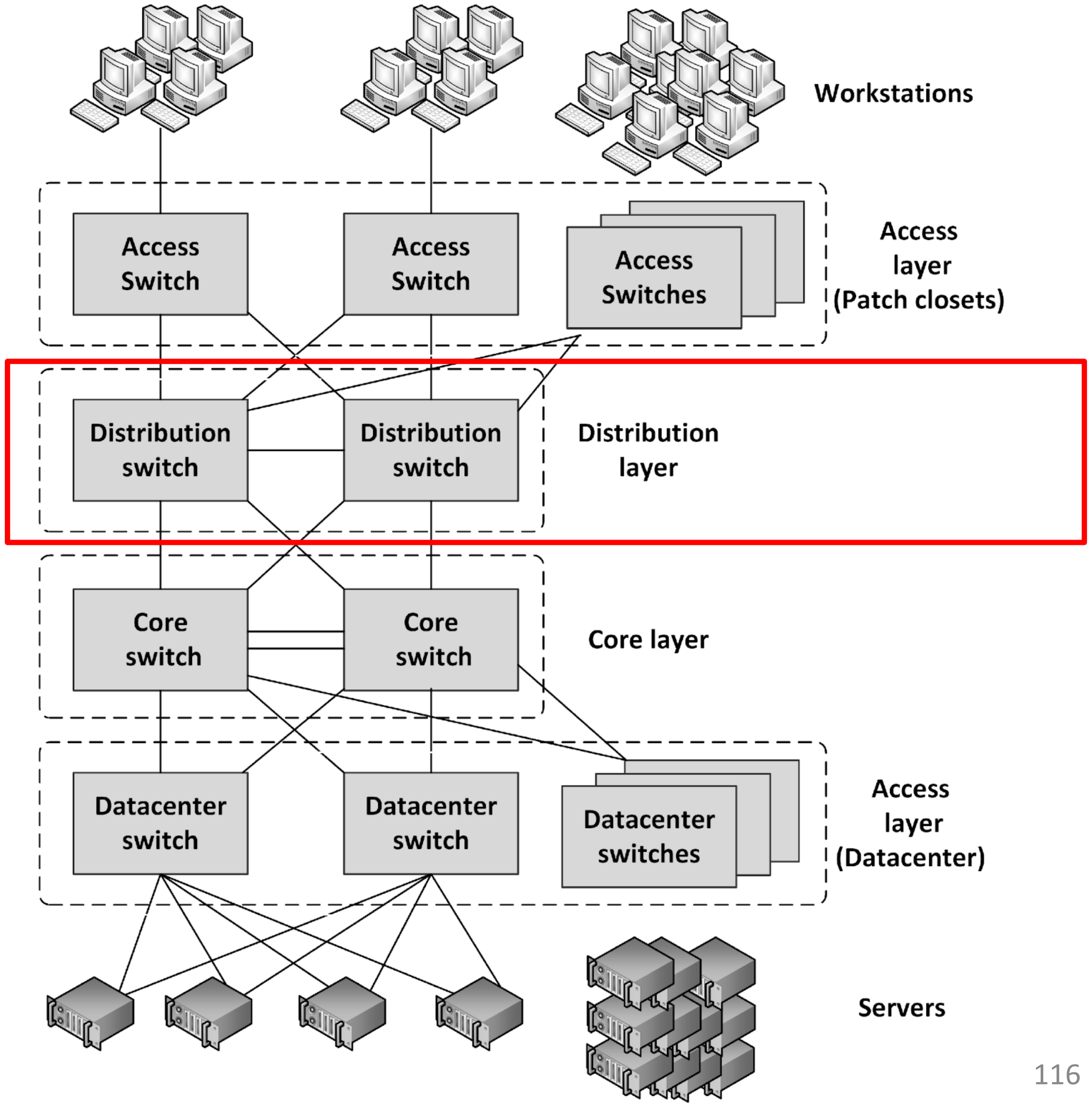
- The center of the network
- The backbone of the network, responsible for high-speed data transfer between different network segments
- It typically uses high-capacity routers and switches.



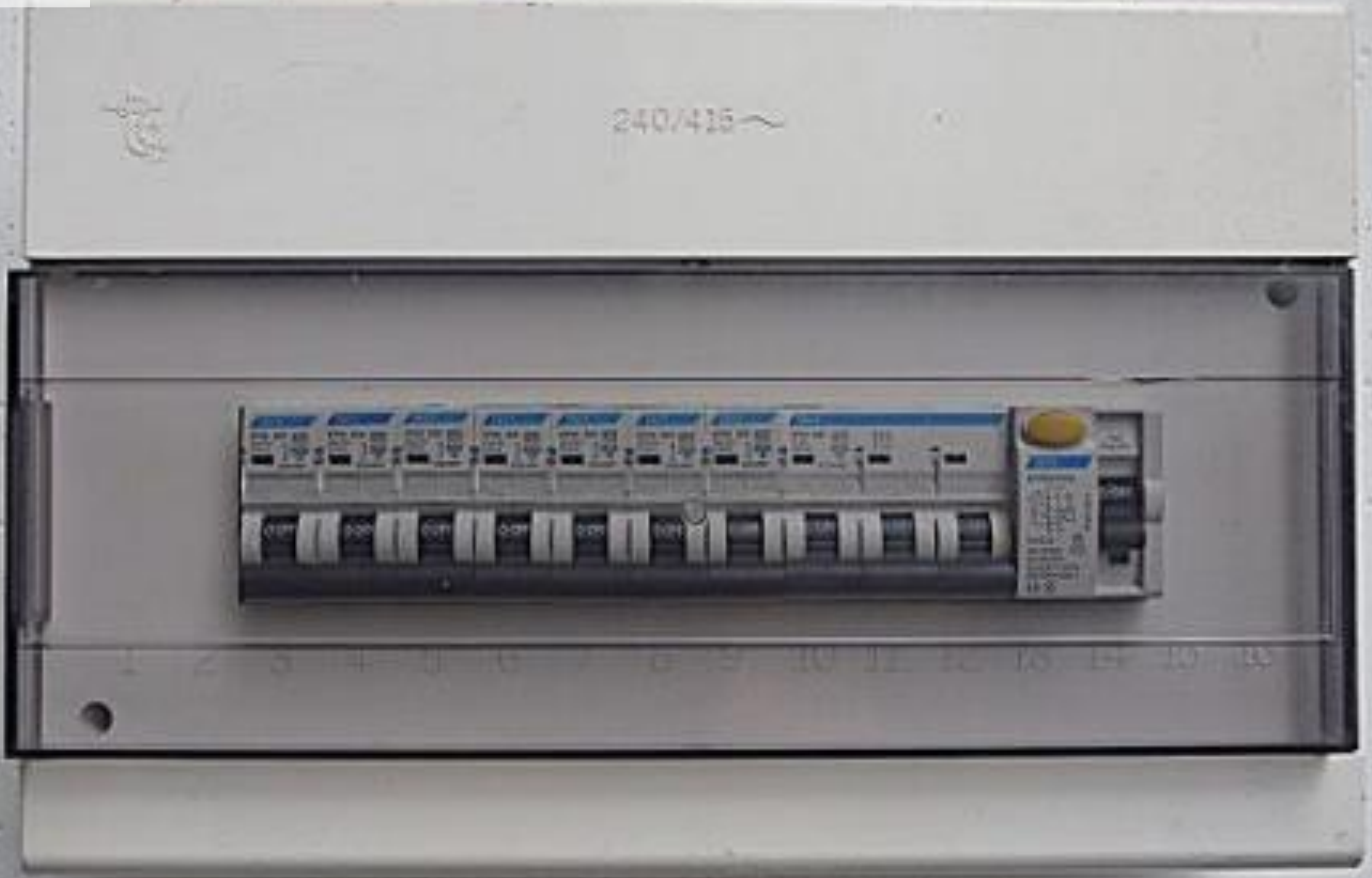
Core Layer

Layered network topology

- **Distribution layer** - an intermediate layer between the core layer in the datacenter and the access switches in the patch closets. It combines the access layer data and sends its combined data to one or two ports on the core switches

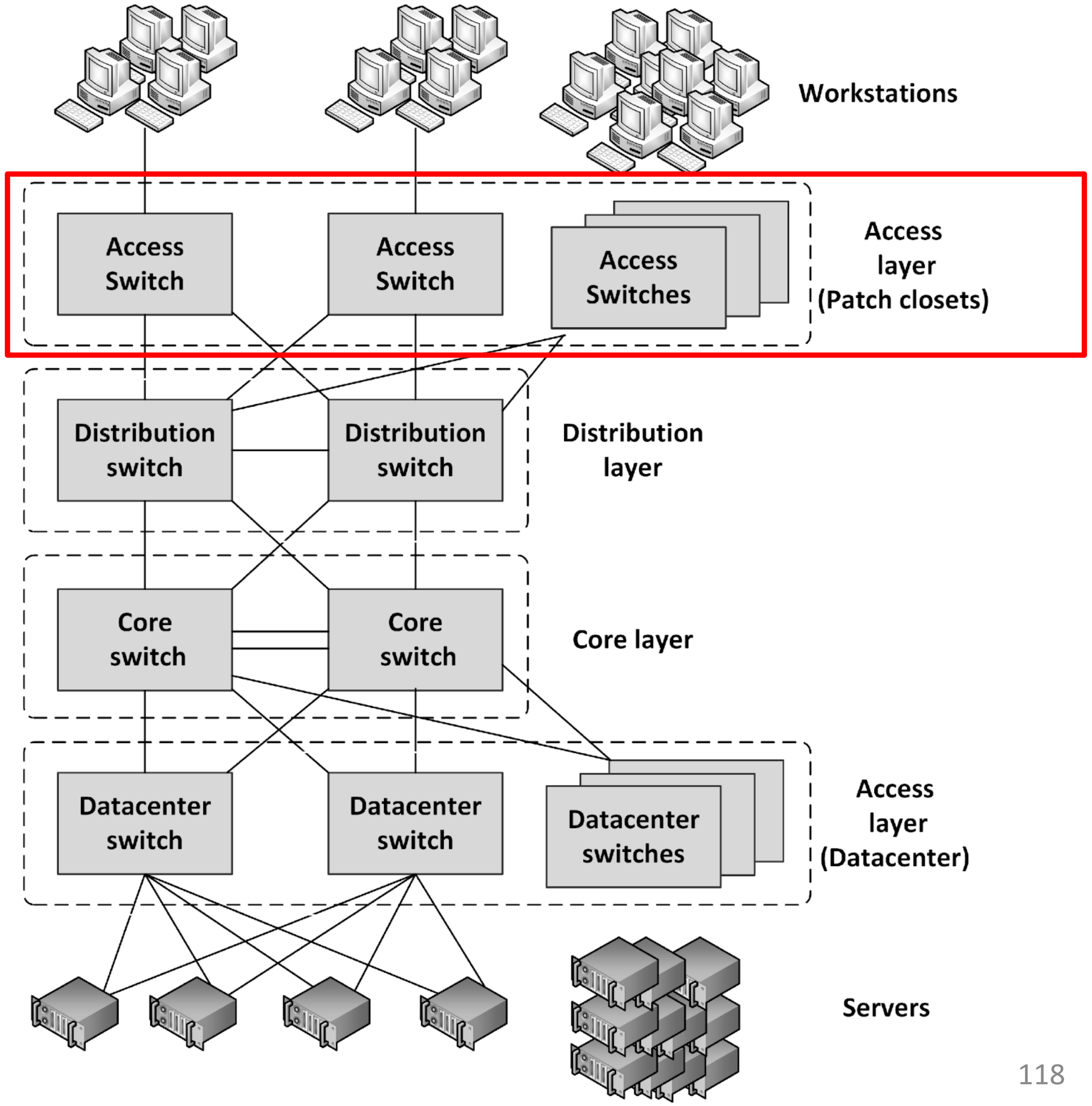


Distribution Layer



Layered network topology

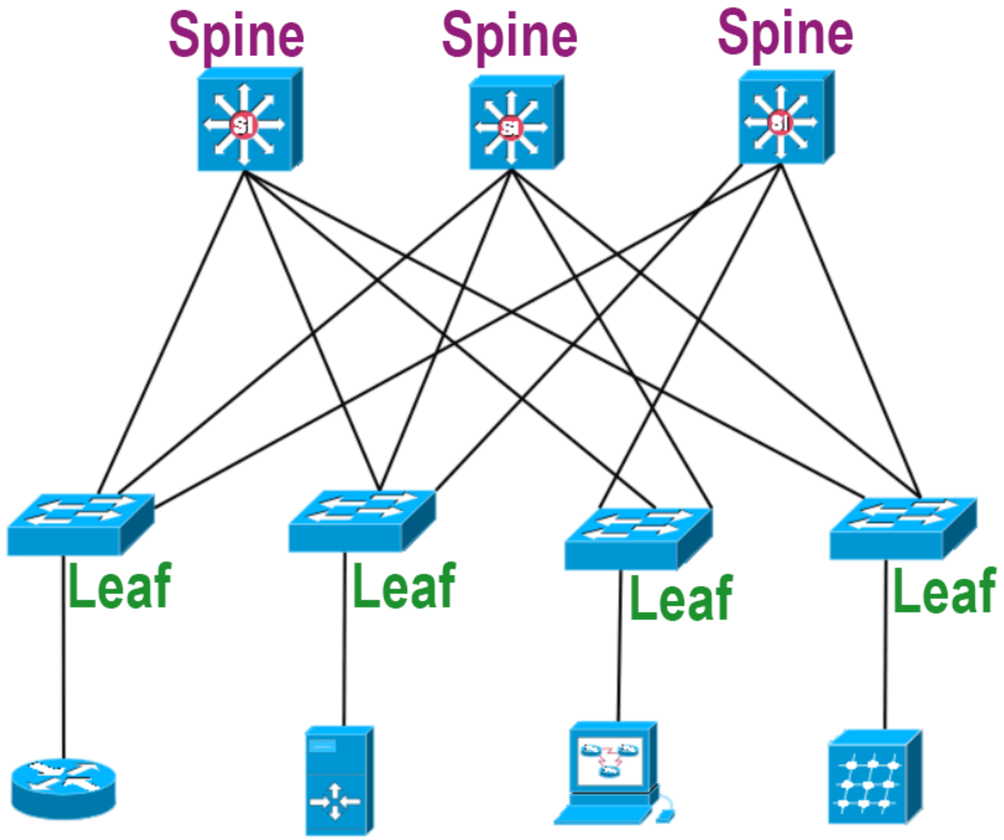
- **Access layer**
  - Connect workstations and servers to the distribution layer
  - For servers, located at the top of the individual server racks or in blade enclosures
  - For workstations, placed in patch closets in various parts of the building



Access Layer

Spine and Leaf topology

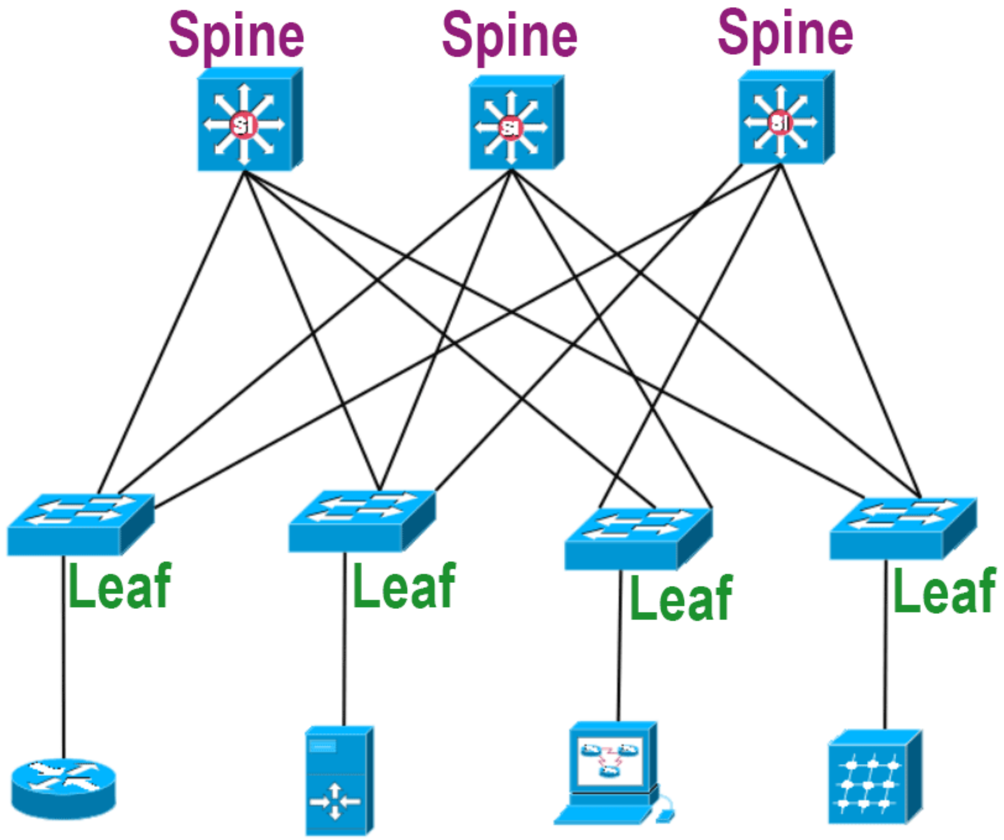
- Spine-Leaf Topology is a modern data center network architecture designed to address the limitations of traditional three-tier hierarchical designs.
- It offers **high scalability**, **performance**, and **resilience**.



Spine and Leaf topology

- In a SDN, a **simple physical network** is used that can be programmed to **act as a complex virtual network**

- Such a network can be organized in a spine and leaf topology

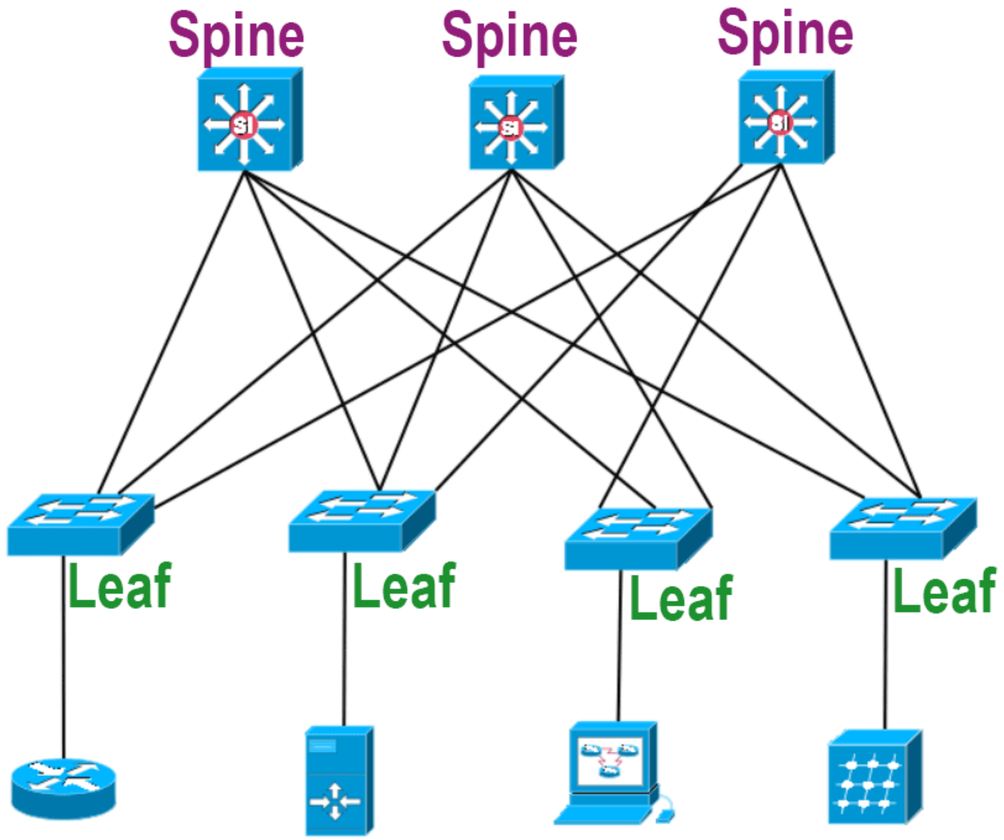


Spine and Leaf topology

Key Components:

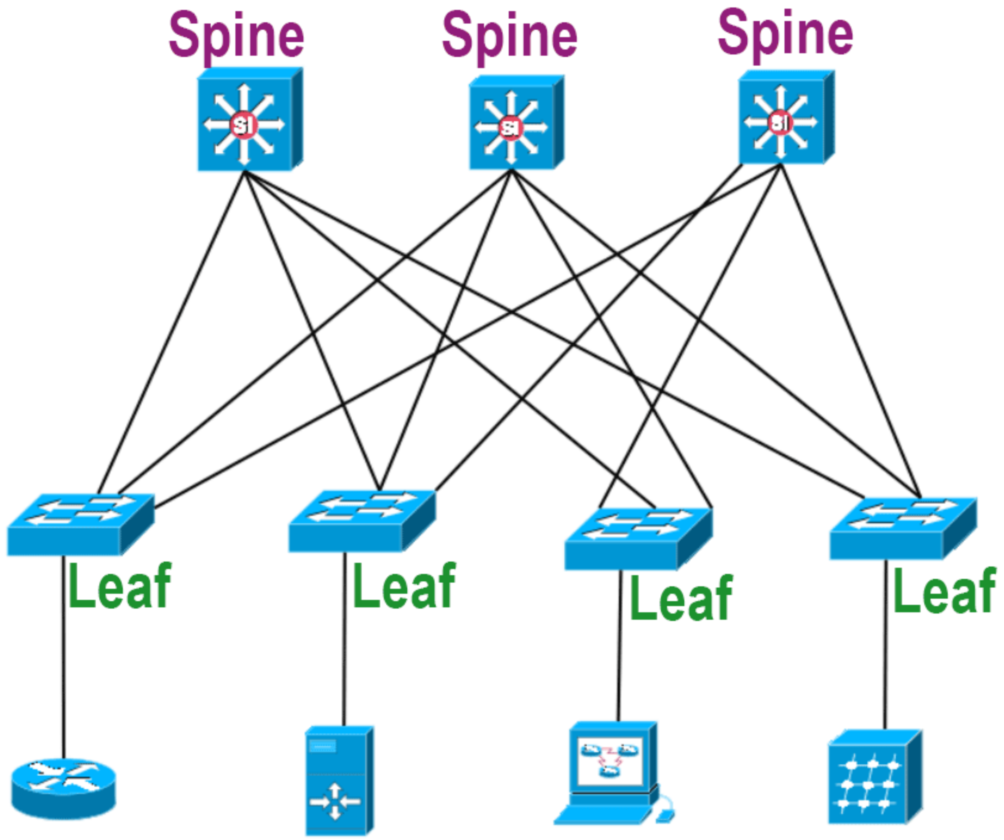
1. Spine Switches
   - Form the **core** of the network.
   - They are interconnected in a **full-mesh topology**, providing redundancy and load balancing.

2. Leaf Switches
   - **Connect to servers** and other **end devices**.
   - Each **leaf** switch **connects to all spine switches**, creating **multiple paths** for traffic.

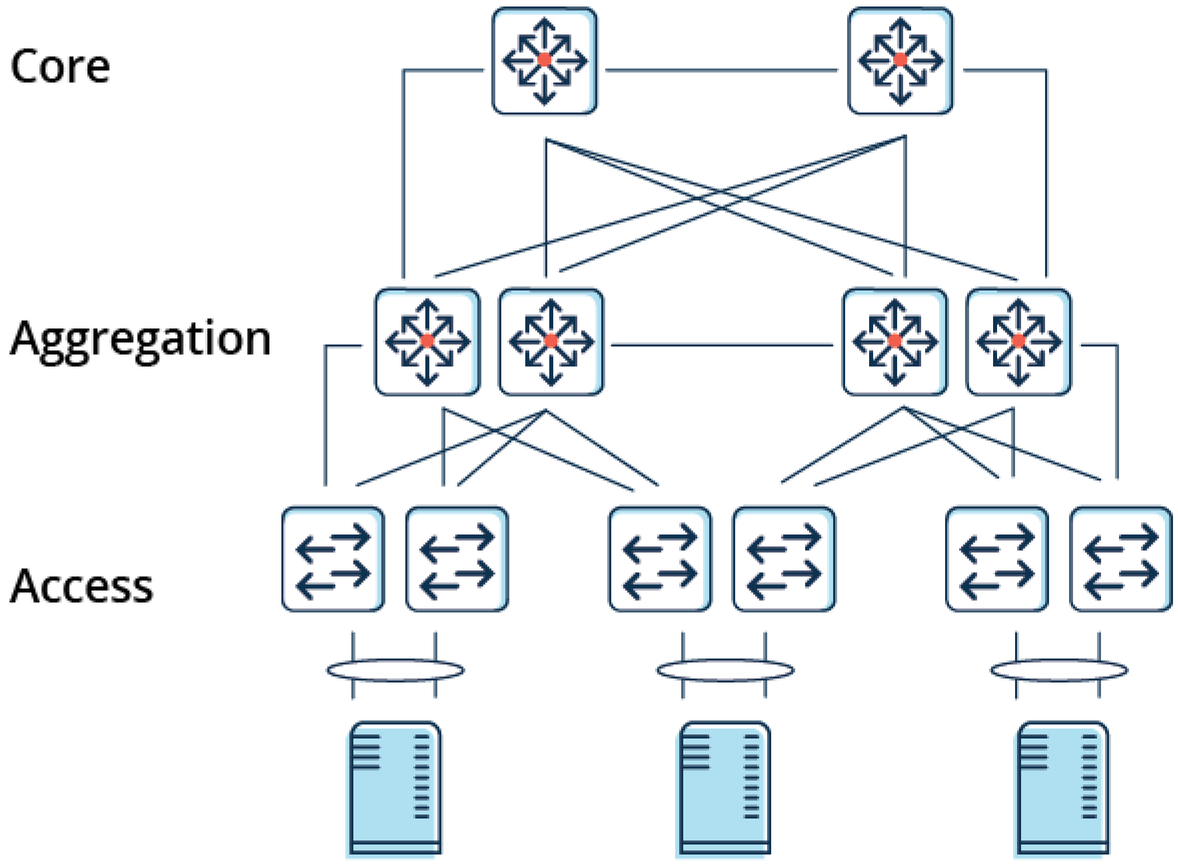


Spine and Leaf topology

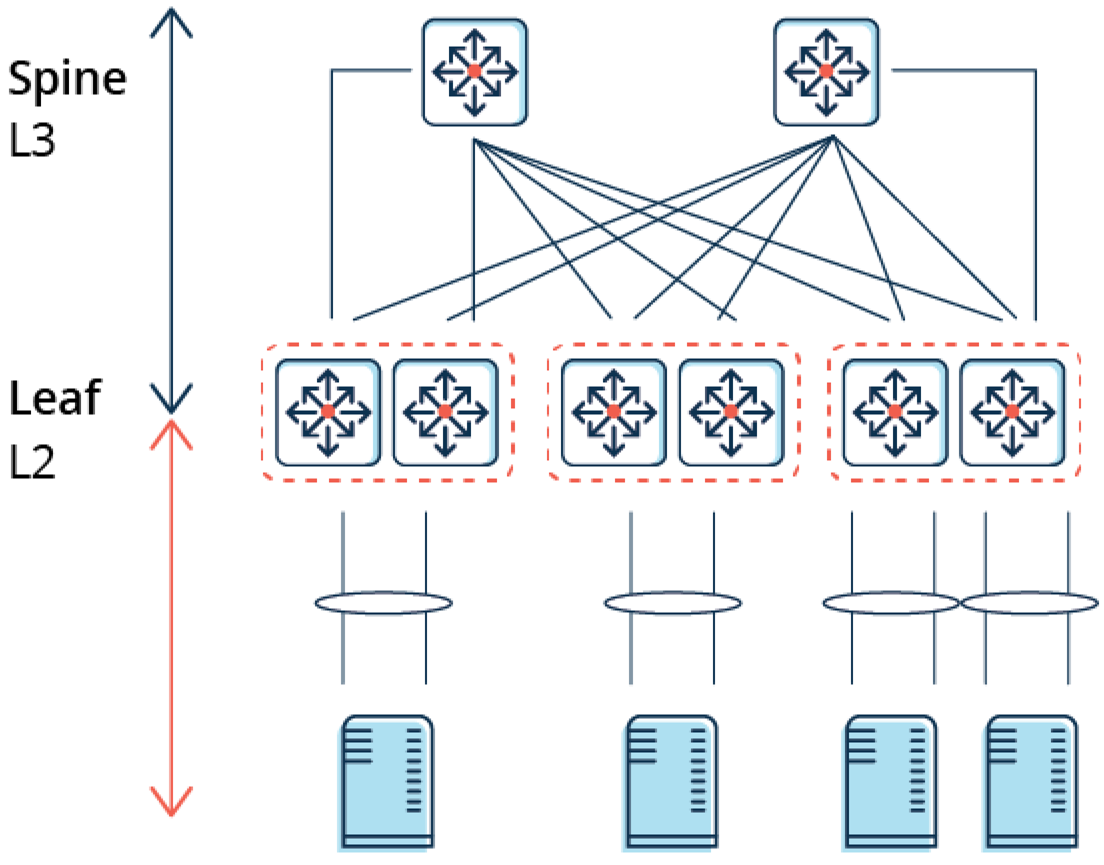
- Characteristics:
  - The spine switches are not interconnected
  - Each leaf switch is connected to all spine switches
  - Each server is connected to two leaf switches
  - The connections between spine and leaf switches typically have ten times the bandwidth of the connectivity between the leaf switches and the servers



Traditional 3-Tier Architecture



2-Tier Spine-Leaf Architecture



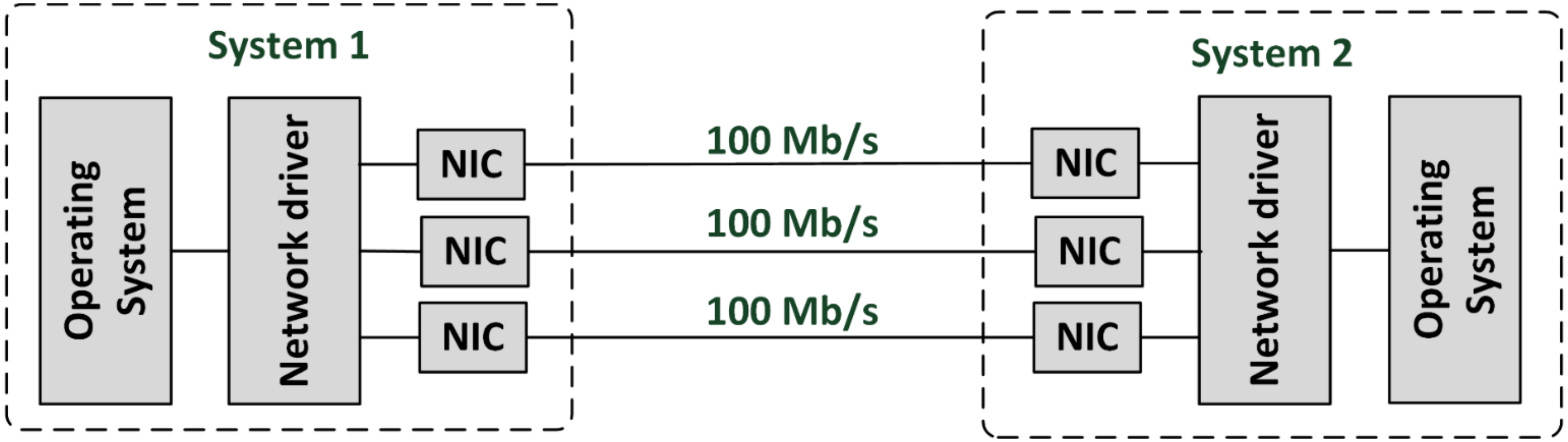

2
2

Spine and Leaf topology

- Benefits:
  - Highly scalable
    - There are no interconnects between the spine switches
  - Simple to scale
    - Just add spine or leaf servers
  - With today’s high density switches, many physical servers can be connected using relatively few switches
  - Each server is always exactly four hops away from every other server
    - Leads to a very predictable latency

Network teaming

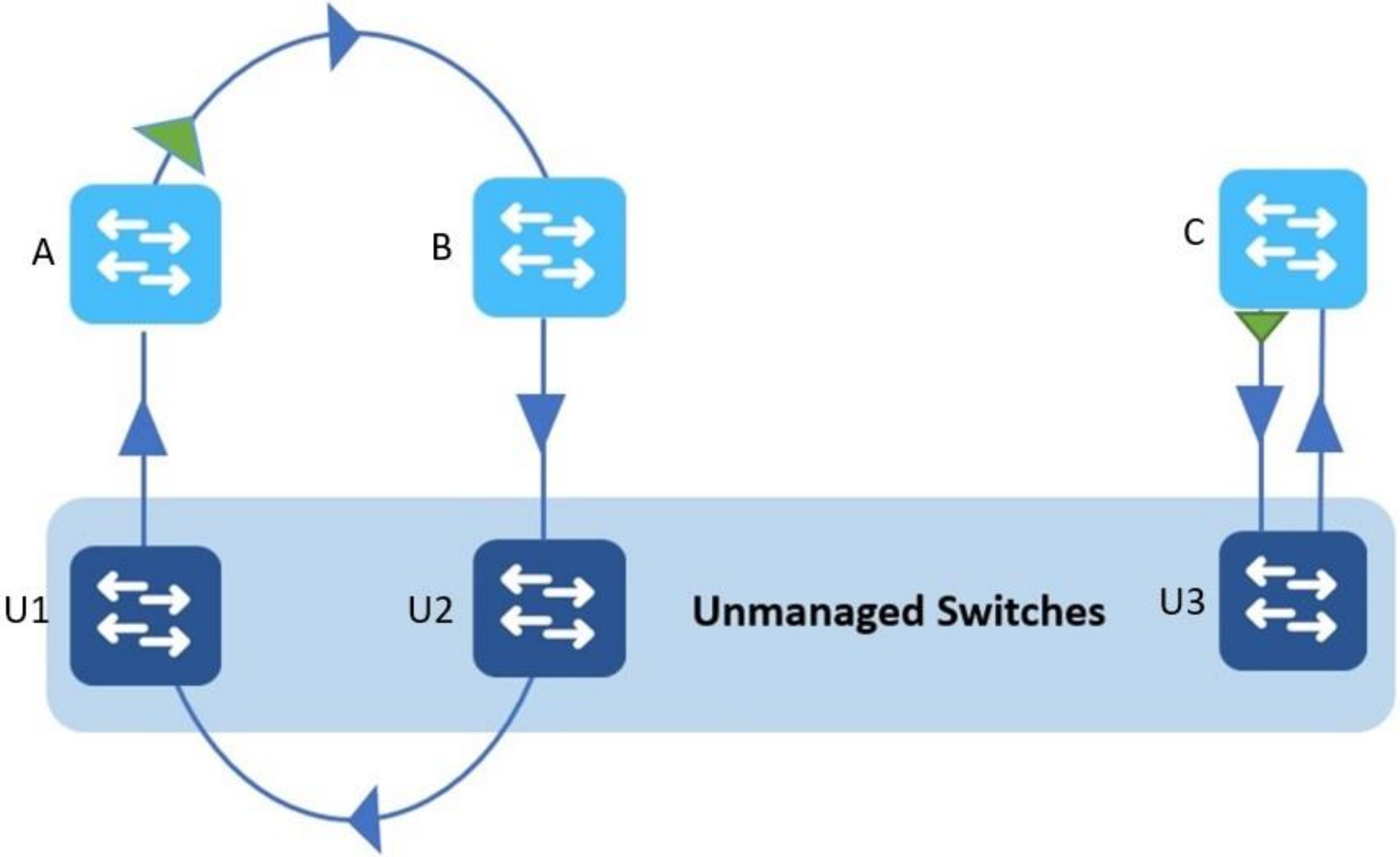
- Network teaming is also known as:
  - Link aggregation
  - Port trunking
  - Network bonding
- Provides a virtual network connection using multiple physical cables for high availability and increased bandwidth



Network teaming

- Network teaming bonds physical NICs together to form a logical network team
  - Sends traffic to the team's destination to all NICs in the team
  - Allows a single NIC, cable, or switch to be unavailable without interrupting traffic

Typical Scenario



Traffic sourced from Switch A to unmanaged Switch U2, is forwarded right back to A through unmanaged Switch U1. Also, traffic from Switch C is looped back to C through U3

Spanning Tree Protocol (STP)

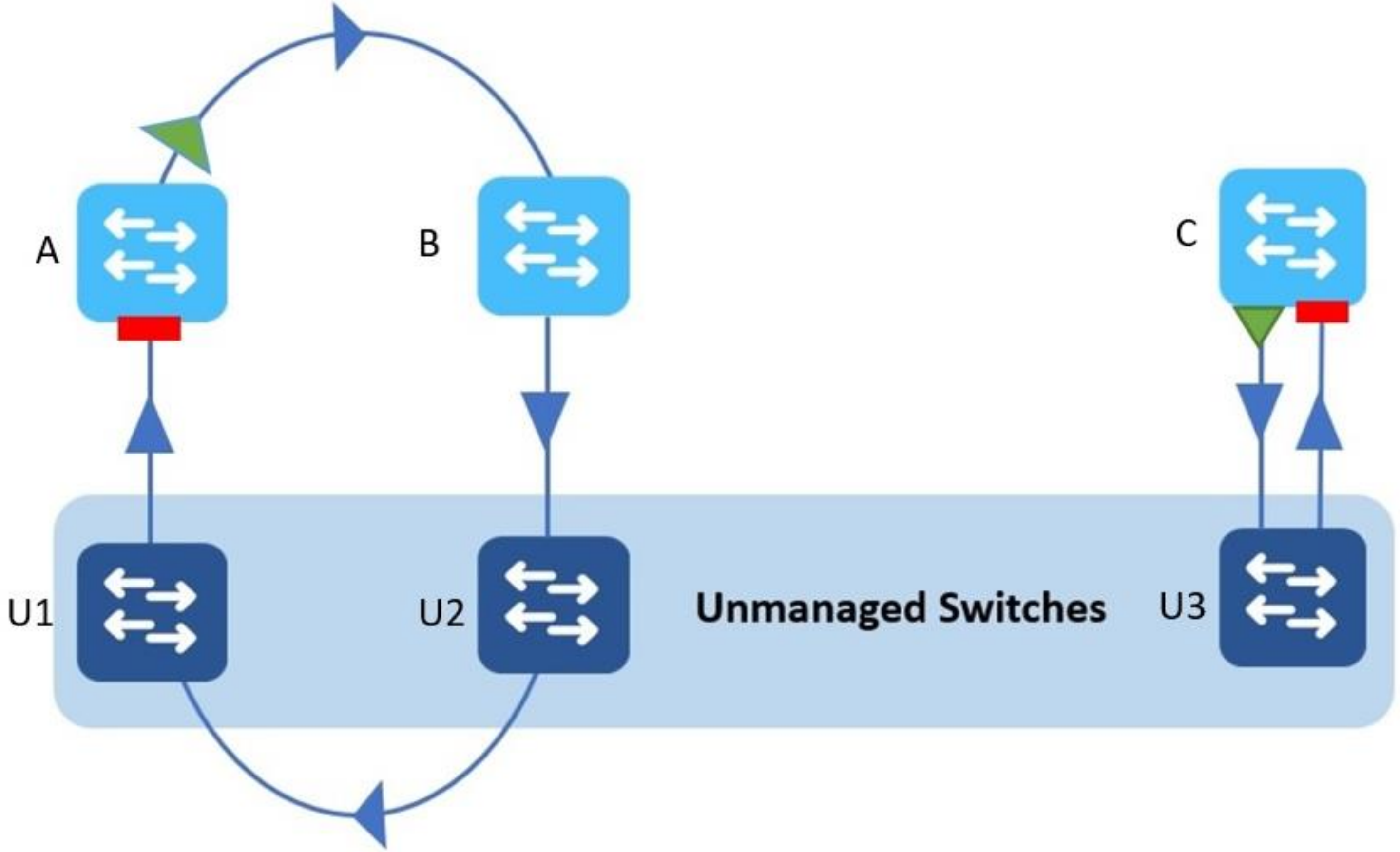
- STP is an Ethernet level protocol that runs on switches

- Guarantees that **only one path is active** between **two network endpoints** at any given time

- **Redundant paths** are automatically **activated** when the **active path experiences problems**

- Ensures **no loops** are created **when redundant paths are available** in the network

Loop Detection Guard



With Loop Detection Guard enabled; when loopback frames return to Switch A, the port is error-disabled

Spanning Tree Protocol (STP)

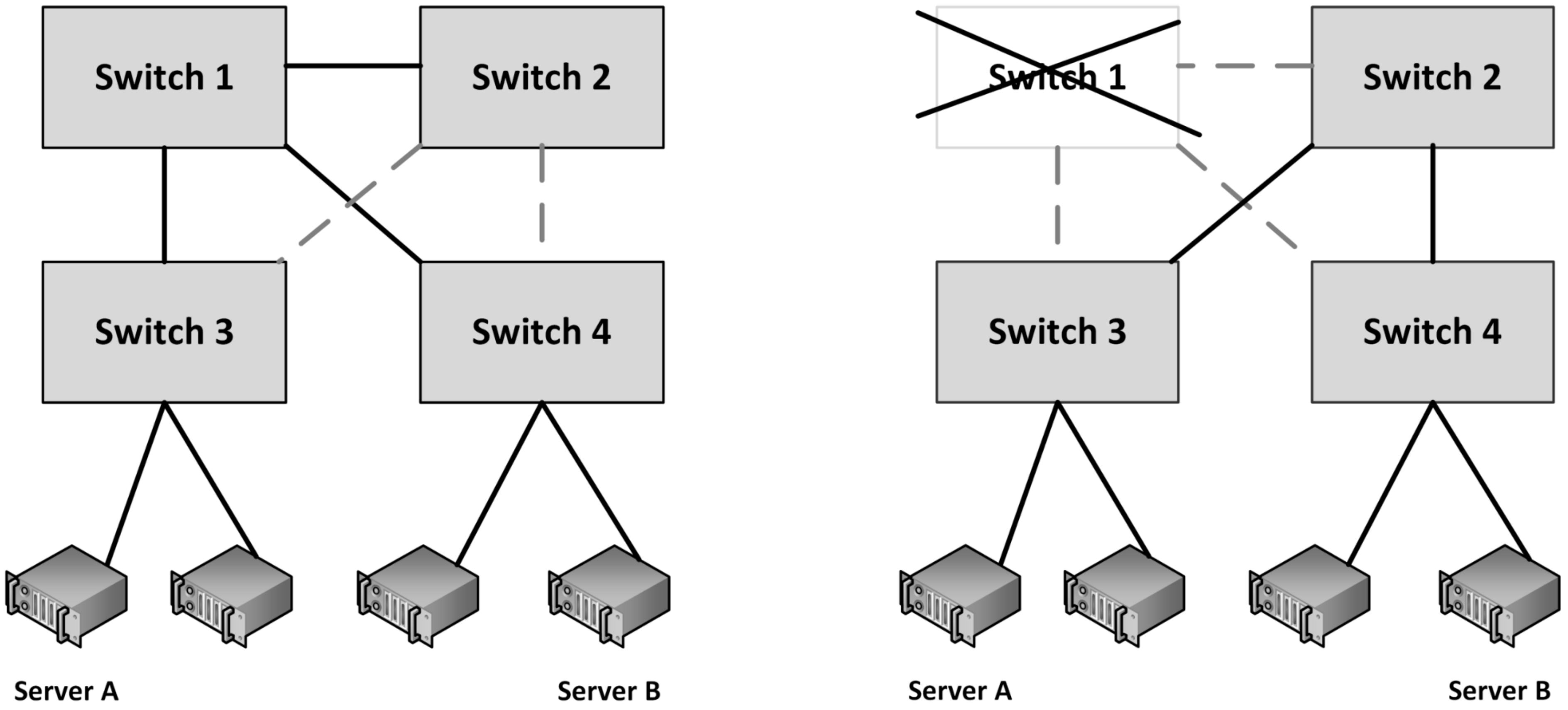
- A disadvantage of using the spanning tree protocol is that it is **not using half of the network links** in a network, since it blocks redundant paths

Spanning Tree Protocol (STP)

- **Shortest Path Bridging** (SPB) allows **all paths** to be **active** simultaneously, enables much larger topologies, supports faster convergence times, and improves efficiency by allowing traffic to be load balanced across all paths

- While STP can take 30 to 60 seconds to respond to a topology change, SPB can respond to changes in less than a second

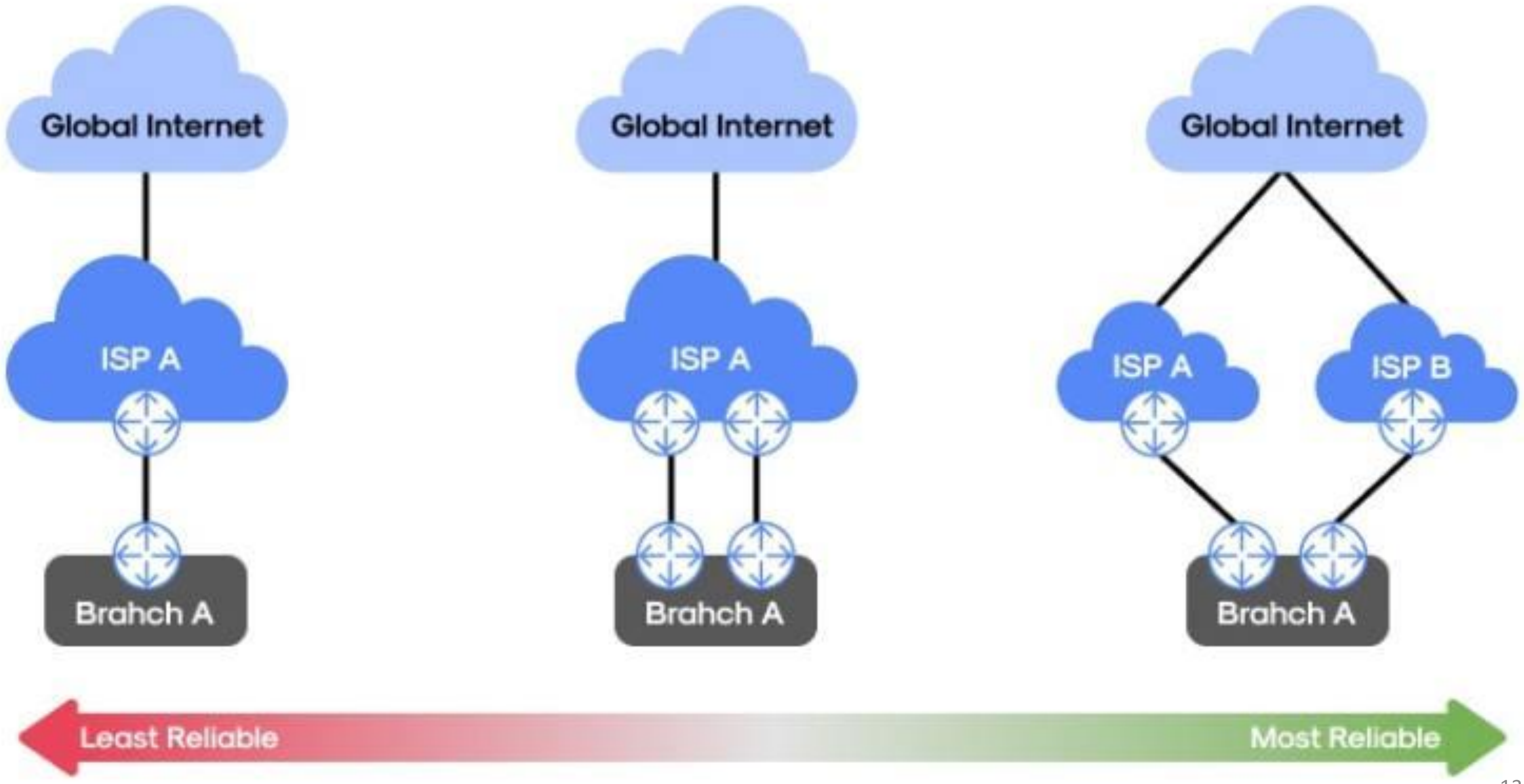
Spanning Tree Protocol



Multihoming

- Connecting a network to two different Internet Service Providers (ISPs) is called multihoming
- Four options for multihoming:
  - Single router with dual links to a single ISP
  - Single router with dual links to two separate ISPs
  - Dual routers, each with its own link to a single ISP
  - Dual routers, each with its own link to a separate ISP

Multi-Homing



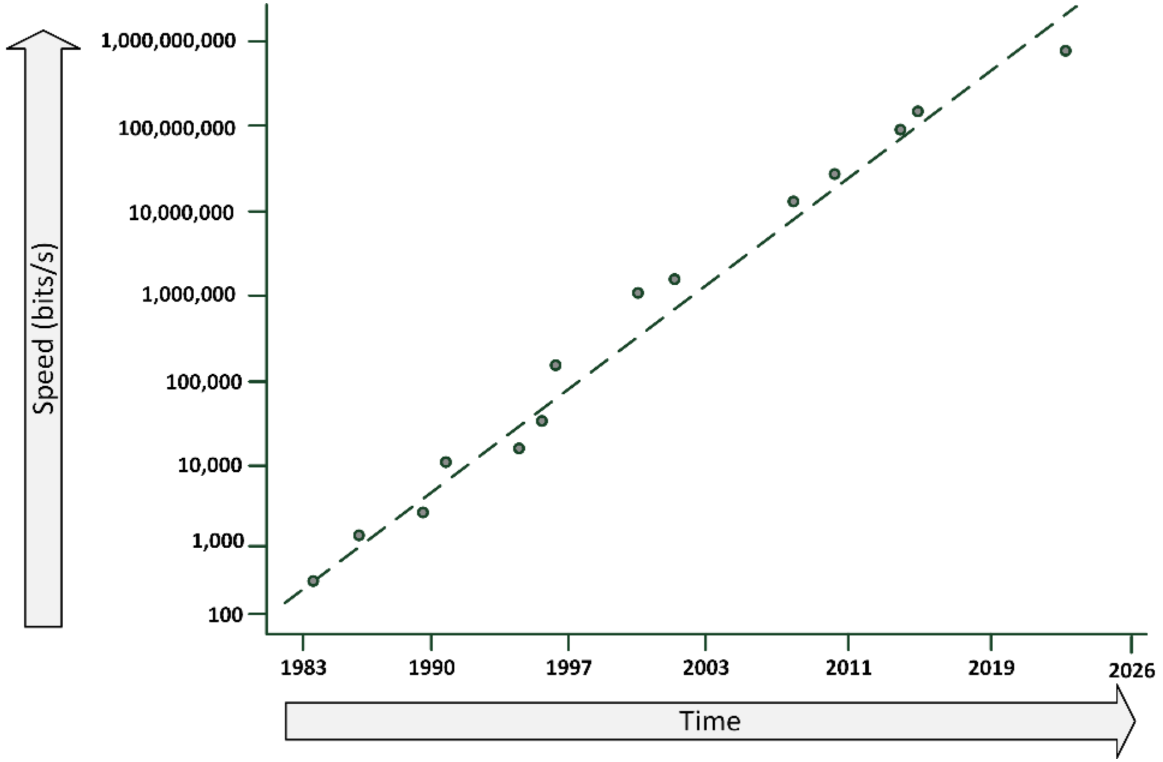
Multihoming

- It is not always guaranteed that multiple network paths actually run on a different set of cables
  - WAN cables are typically installed alongside highways and railway tracks
  - Cables are used by multiple carrier providers

Network performance

Nielsen’s law

- Network connection speeds for high-end home users increase 50% per year, they double every 21 months
- Regular bandwidths should be 2 Gbit/s in 2026, still for a price of about MYR 220 per month



Please note that the vertical scale is logarithmic instead of linear

Throughput and bandwidth

- **Throughput** is the **amount of data** that is **transferred** through the network during a **specific time interval**
- Throughput is limited by the available **bandwidth**
- When an application requires more throughput than a network connection can deliver:
  - Queues in the network components temporarily buffer data
  - Buffered data is sent as soon as the network connection is free again
  - When more data arrives than the queues can store in the buffer, packet loss occurs


Throughput
Bandwidth

Latency

- Latency is defined as the **time** from the **start** of packet **transmission** to the start of **packet reception**

- Latency is dependent on:
  - The physical distance a packet has to travel
  - The number of switches and routers the packet has to pass

- Rules of thumb:
  - 6 ms latency per 100 km
  - WANs: Each switch in the path adds 10 ms to the one-way delay
  - LANs: add 1 ms for each switch

Latency

- One-way latency: the time from the source sending a packet to the destination receiving it

- Round-trip latency: the one-way latency from source to destination plus the one-way latency from the destination back to the source

- “ping” can be used to measure round-trip latency

Quality of Service (QoS)

- Quality of service (QoS) is the ability to provide **different** data flow **priority** to different applications, users, or types of data

- QoS allows **better service** to certain **important data flows** compared to less important data flows

- QoS is mainly used for **real-time applications** like video and audio streams and VoIP telephony

Quality of Service (QoS)

- Four basic ways to implement QoS:

  1. Congestion management

  2. Queue management

  3. Link efficiency

  4. Traffic shaping

Quality of Service (QoS)

1. Congestion management

- Defines what must be done if the amount of data to be sent exceeds the bandwidth of the network link
- Packets can either be dropped or queued

Quality of Service (QoS)

2. Queue management

- When queues are full, packets will be dropped
- Queue management defines criteria for dropping packets that are of lower priority before dropping higher priority packets

Quality of Service (QoS)

3. Link efficiency

- Ensures the link is used in an optimized way

- For instance by fragmenting large packets with a low QoS, allowing packets with a high QoS to be sent between the fragments of low QoS packets

Quality of Service (QoS)

4. Traffic shaping

- Limiting the full bandwidth of streams with a low QoS to benefit streams with a high QoS
- High QoS streams have a reserved amount of bandwidth

WAN link compression

- Data compression reduces the size of data before it is transmitted over a WAN connection

- WAN acceleration appliances:
  - Provide compression
  - Perform some caching of regularly used data at remote sites

Network security

Network encryption

- All data that traverses the network should be encrypted
- Encrypting data on the network is called encrypting data in transit (as opposed to encrypting data at rest)
- Data can be encrypted between two endpoints, such as a server and a client. This is called end-to-end encryption. An example is HTTPS for traffic from a web server to a web browser
- Even when using end to end encryption, network traffic must always be encrypted
- Encryption is often a feature of the network components in the datacenter

Firewalls

- Firewalls separate two or more LAN or WAN segments for security reasons
- Firewalls block all unpermitted network traffic between network segments
- Permitted traffic must be explicitly enabled by configuring the firewall to allow it

Firewalls

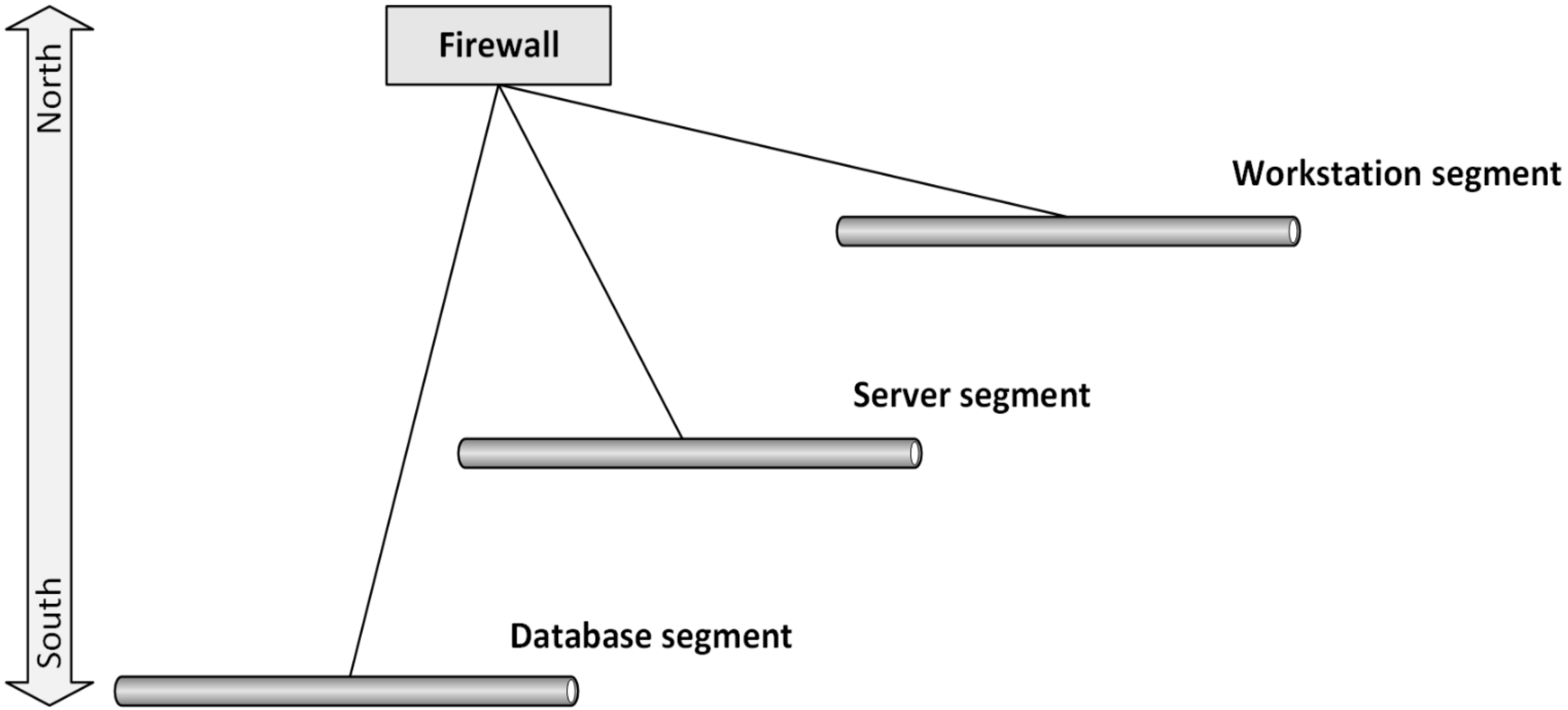
- Firewalls can be implemented:
  - In hardware appliances
  - As an application on physical servers
  - In virtual machines
- Host based firewalls
  - Protect a server or end user computer against network based attacks
  - Part of the operating system

Firewalls

- Firewalls use one or more of the following methods to control traffic:
  - **Packet filtering**
    - Data packets are analyzed using preconfigured filters
    - This functionality is almost always available on routers and most operating systems
  - **Proxy** (also known as application layer firewalls or API gateways)
    - A proxy terminates the session on the application level on behalf of the server (proxy) or the client (reverse proxy) and creates a new session to the client or server
  - **Stateful inspection**
    - Inspects the placement of each individual packet within a packet stream
    - Maintains records of all connections passing through the firewall and determines whether a packet is the start of a new connection, part of an existing connection, or is an invalid packet

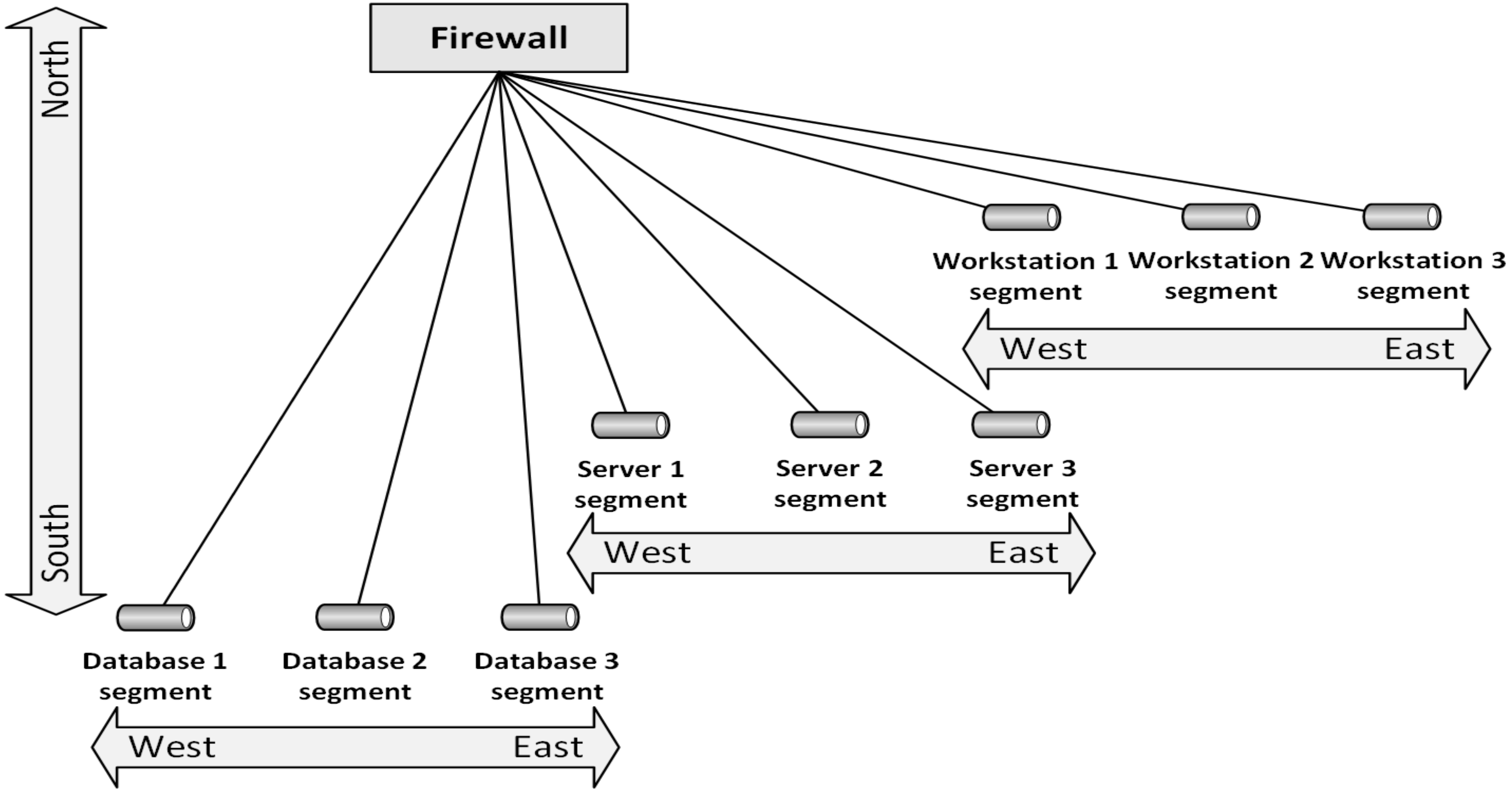
Network segmentation

Segmentation



Network segmentation

Micro segmentation



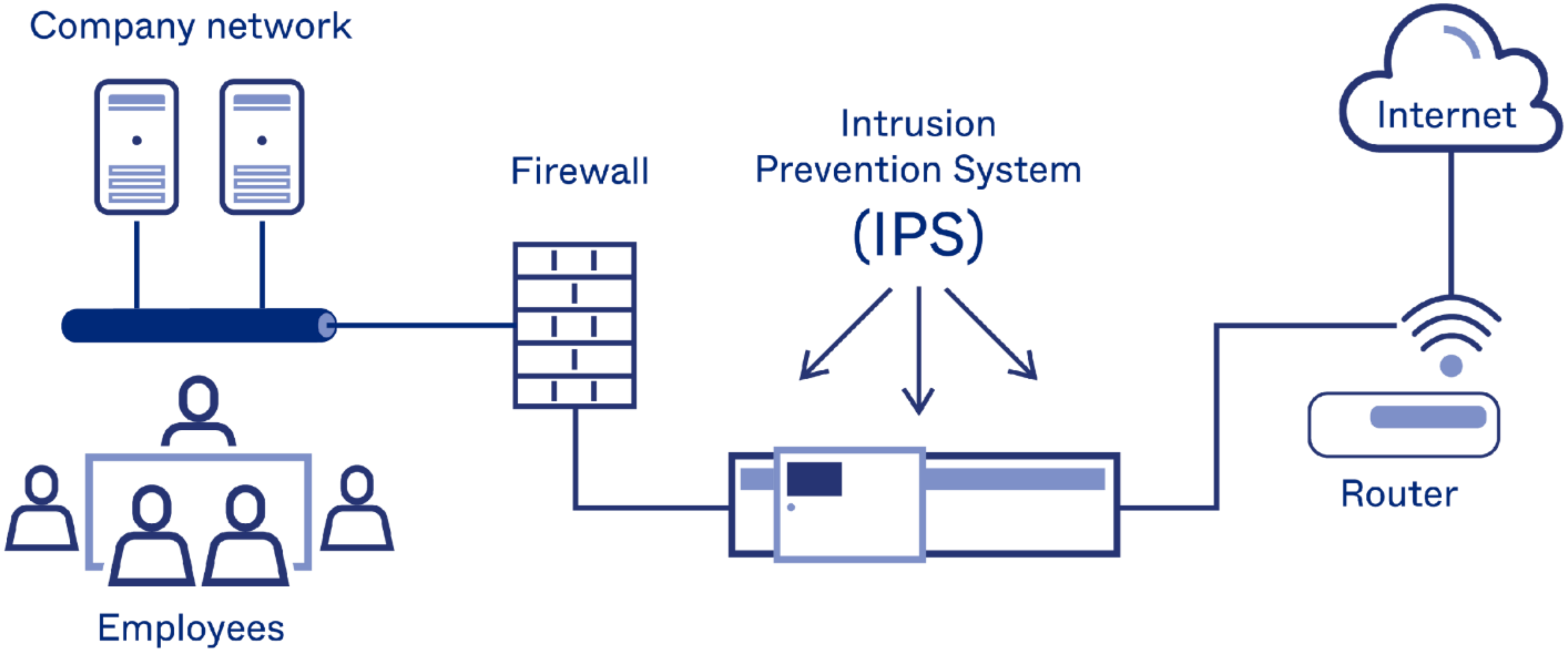
IDS/IPS

- An **Intrusion Detection System** (IDS) or **Intrusion Prevention System** (IPS) detects and – if possible – prevents activities that compromise system security, or are a hacking attempt

- An IDS/IPS monitors for suspicious activity and alerts the systems manager when these activities are detected

- An IPS can stop attacks by changing firewall rules on the fly

IDS/IPS



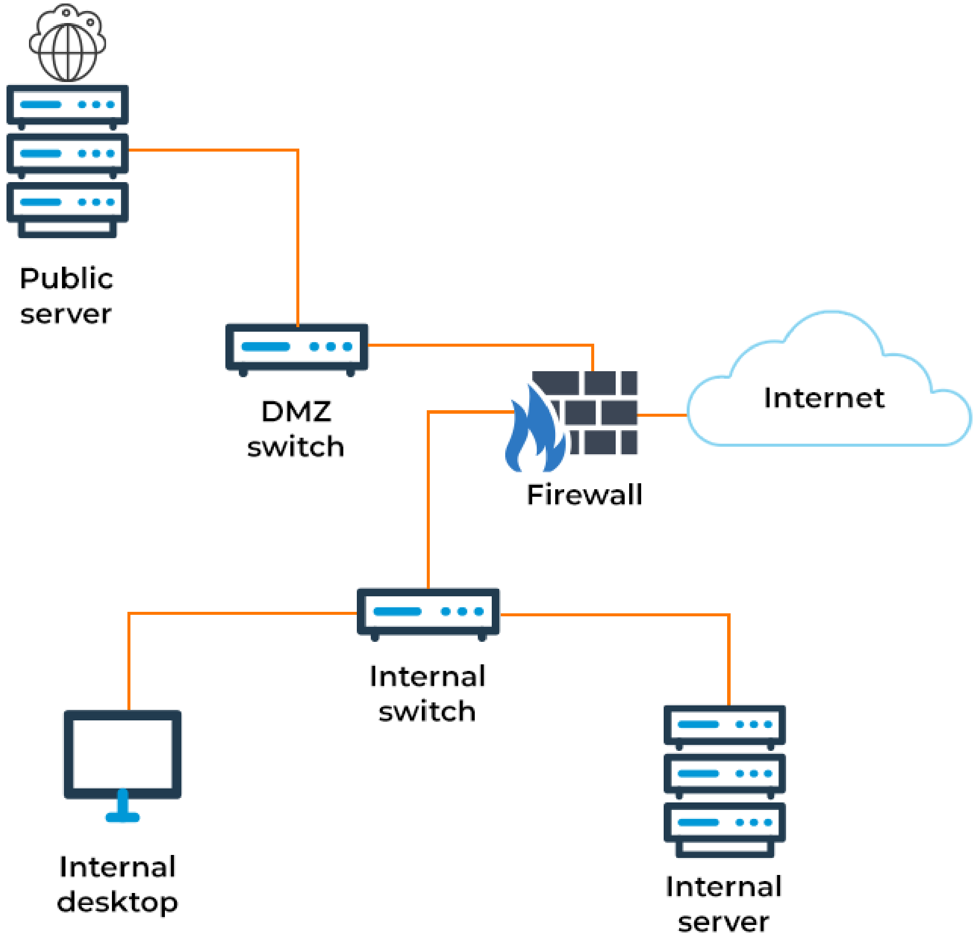
Company network
Firewall
Intrusion
Prevention System
(IPS)
Internet
Router
Employees

IDS/IPS

- Two types of IDS/IPS systems:
  - A **Network-based IDS (NIDS)** is placed at a **strategic point in the network**
    - Monitors traffic to and from all devices on that network
    - The NIDS is not part of the network flow, but just “looks at it”, to avoid detection of the NIDS by hackers
  - A **Host-based IDS (HIDS)** runs on **individual servers** or network devices
    - It monitors the network traffic of that device
    - It also monitors user behavior and the alteration of critical (system) files

DMZ

- DMZ is short for **De-Militarized Zone**, also known as screened subnet, or the Perimeter Network
- A DMZ is a network that serves as a buffer between a secure protected internal network and the insecure internet

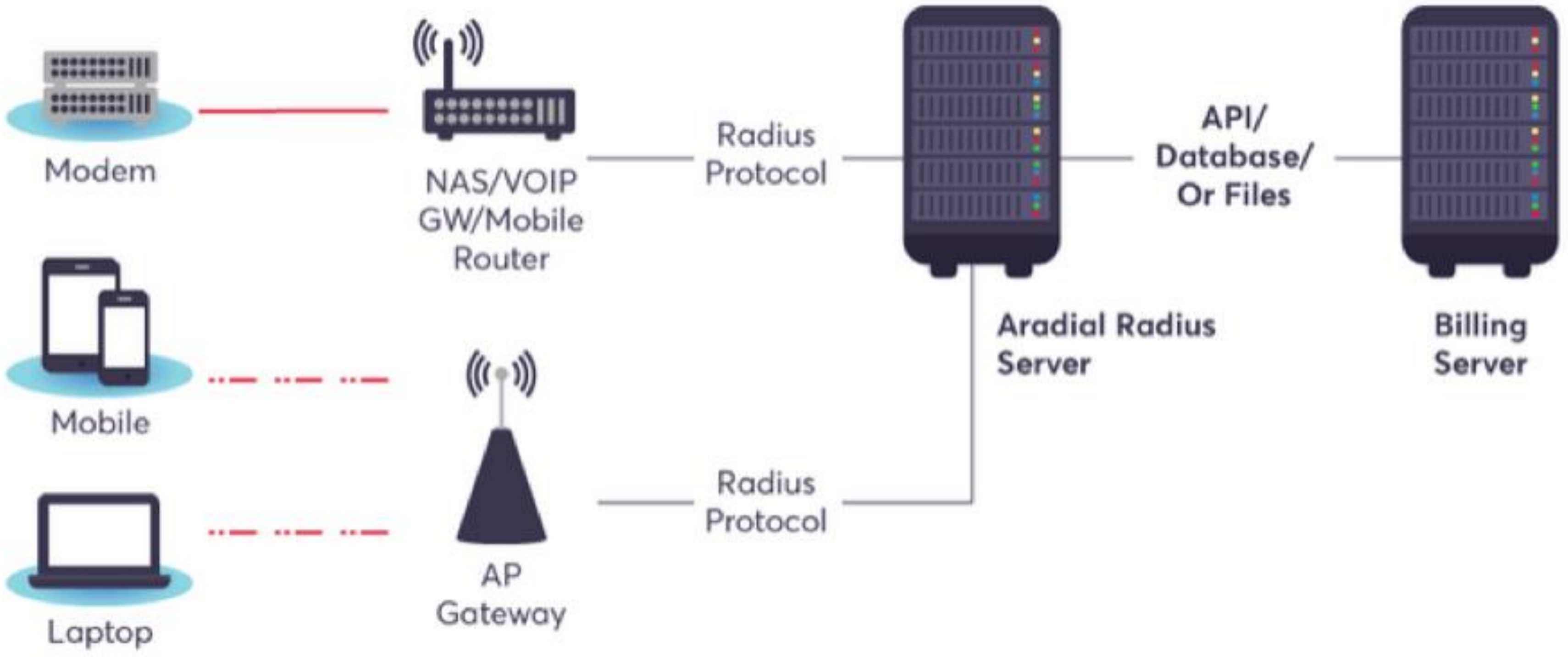


RADIUS

- **Remote Authentication Dial In User Service** (RADIUS) is a networking protocol that provides centralized user and authorization management for network devices
  - Routers
  - Modem servers
  - Switches
  - VPN routers
  - Wireless network access points

RADIUS

- RADIUS
  - Authenticates users or devices before granting them access to a network
  - Authorizes users or devices for certain network services



Conclusion

- Explain the networking building blocks
- Explain network virtualization
- Explain network availability
- Explain network performance
- Explain network security